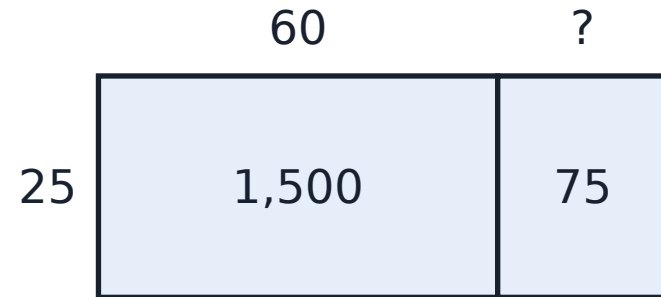


The area model shows
 $1,575 \div 25$.

What is the quotient?



Quotient: _____

True or false?

An area model for $2,016 \div 48$ can use parts with areas 1,920 and 96, so the quotient is $40 + 2 = 42$.

- A.** True
- B.** False

Which equations show $1,410 \div 30$?

A. $1,410 - 30 = 1,380$

B. $30 \times 40 = 1,200$; $30 \times 7 = 210$; $40 + 7 = 47$

C. $30 \times 47 = 1,140$

D. $30 \times 4 = 120$; $30 \times 7 = 210$; $4 + 7 = 11$

Divide.

$$2,262 \div 39$$

Quotient: _____

True or false?

An area model for $1,344 \div 32$ with parts 32×40 and 32×3 shows a quotient of 43.

- A. True
- B. False

SET 26 • 4.NBT.B.6

FORWARD 2

Connect the standard division algorithm to partial quotients • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$6,912 \div 24$$

Quotient: _____

In the standard algorithm for $4,284 \div 12$, the first step uses $12 \times 300 = 3,600$.

What does the 300 stand for?

- A.** Part of the quotient
- B.** The remainder
- C.** Part of the divisor
- D.** The dividend

True or false?

The steps of the standard algorithm for $3,705 \div 15$ match the partial quotients $200 + 40 + 7$.

A. True

B. False

Divide.

$$10,488 \div 76$$

Quotient: _____

True or false?

$$5,616 \div 52 = 118$$

A. True

B. False

Explain why fractions are equivalent using visual fraction models

MAIN	Grade 4 main question	4.NF.A.1
BACKWARD 1	Recognize equivalent fractions with a visual model	3.NF.A.3.b
BACKWARD 2	Equivalent fractions are at the same point on a number line	3.NF.A.3.a
FORWARD 1	Use equivalent fractions to add and subtract with unlike denomina...	5.NF.A.1
FORWARD 2	Find equivalent ratios	6.RP.A.3.a

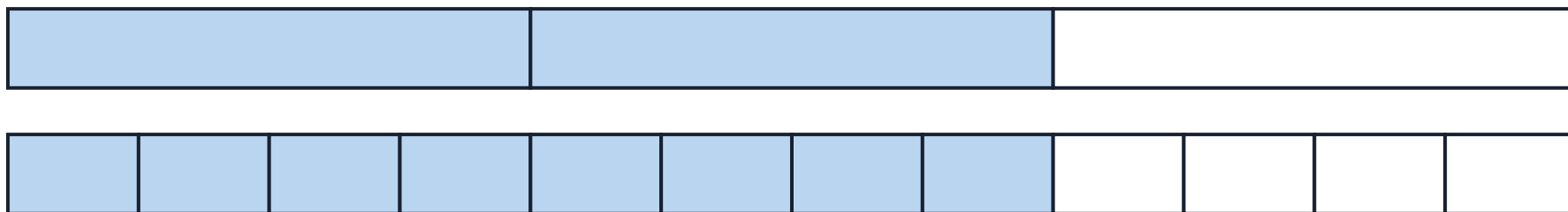
SET 27 • 4.NF.A.1

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

The bars show $\frac{2}{3}$ and $\frac{8}{12}$ of the same whole.

Explain how the model shows that $\frac{2}{3} = \frac{8}{12}$.



Explanation: _____

A bar shows $\frac{3}{5}$ shaded. Each fifth is cut into 2 equal parts.

Which fraction now names the shaded part?

A. $\frac{6}{10}$

B. $\frac{3}{10}$

C. $\frac{6}{5}$

D. $\frac{5}{10}$

True or false?

$\frac{3}{4} = \frac{9}{12}$ because the numerator and the denominator were both multiplied by 3.

A. True

B. False

The bar shows $\frac{1}{2}$ shaded.

Draw lines to split each part into 3 equal parts. Then write the equivalent fraction.



Equivalent fraction: _____

True or false?

$\frac{2}{5} = \frac{4}{5}$ because the numerator was multiplied by 2.

A. True

B. False

SET 27 • 4.NF.A.1

BACKWARD 1

Recognize equivalent fractions with a visual model • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.b

True or false?

$$\frac{1}{2} = \frac{2}{4}$$

A. True

B. False

The bars show the same whole.

What fraction of the second bar names the same amount as $\frac{1}{3}$?



Fraction: _____

Which fraction is equivalent to $\frac{3}{4}$?

A. $\frac{3}{8}$

B. $\frac{4}{6}$

C. $\frac{6}{8}$

D. $\frac{4}{3}$

Find the missing number.

$$\frac{1}{4} = \frac{?}{8}$$

Missing number: _____

True or false?

$$\frac{2}{3} = \frac{2}{6}$$

A. True

B. False

SET 27 • 4.NF.A.1

BACKWARD 2

Equivalent fractions are at the same point on a number line • Grade 3 •
Number & Operations–Fractions • 3.NF.A.3.a

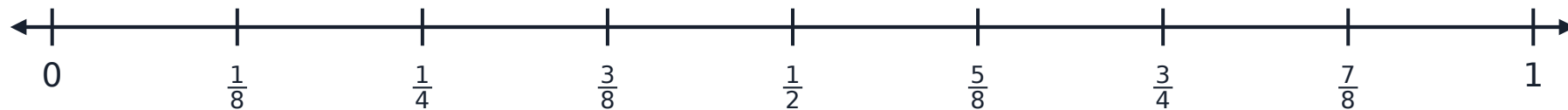
True or false?

$\frac{1}{2}$ and $\frac{2}{4}$ are at the same point on a number line.

A. True

B. False

Which fraction is at the same point on the number line as $\frac{3}{4}$?



A. $\frac{7}{8}$

B. $\frac{3}{8}$

C. $\frac{4}{8}$

D. $\frac{6}{8}$

Name a fraction in eighths that is at the same point as $\frac{1}{2}$.

Fraction: _____

True or false?

$\frac{1}{3}$ and $\frac{2}{6}$ are at different points on a number line.

A. True

B. False

What fraction in sixths is at the same point as $\frac{2}{3}$?

Fraction: _____

SET 27 • 4.NF.A.1

FORWARD 1

Use equivalent fractions to add and subtract with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{1}{3} + \frac{1}{4}$$

Sum: _____

True or false?

$$\frac{2}{5} + \frac{3}{10} = \frac{7}{10}$$

A. True

B. False

Add.

$$\frac{1}{2} + \frac{3}{8}$$

A. $\frac{7}{8}$

B. $\frac{4}{8}$

C. $\frac{3}{16}$

D. $\frac{4}{10}$

Subtract.

$$\frac{5}{6} - \frac{1}{4}$$

Difference: _____

True or false?

$$\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$$

A. True

B. False

SET 27 • 4.NF.A.1

FORWARD 2

Find equivalent ratios • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

Find the missing value.

$$2 : 3 = 8 : ?$$

Missing value: _____

True or false?

3 : 4 and 9 : 12 are equivalent ratios.

A. True

B. False

Which ratio is equivalent to 5 : 2?

- A.** 5 : 4
- B.** 7 : 4
- C.** 15 : 6
- D.** 10 : 2

The table shows equivalent ratios.

Find the missing value.

Cups of flour	2	4	6
Cups of milk	3	6	?

Missing value: _____

True or false?

4 : 6 and 6 : 8 are equivalent ratios.

A. True

B. False

Generate equivalent fractions

MAIN	Grade 4 main question	4.NF.A.1
BACKWARD 1	Generate simple equivalent fractions	3.NF.A.3.b
BACKWARD 2	Whole numbers written as fractions	3.NF.A.3.c
FORWARD 1	Explain equivalent fractions as multiplying by a fraction equal to 1	5.NF.B.5.b
FORWARD 2	Write fractions as percents	6.RP.A.3.c

SET 28 • 4.NF.A.1

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{3}{5} = \frac{?}{10}$$

Missing number: _____

Find the missing number.

$$\frac{8}{12} = \frac{2}{?}$$

Missing number: _____

Which fraction is equivalent to $\frac{4}{6}$?

A. $\frac{2}{6}$

B. $\frac{6}{8}$

C. $\frac{8}{12}$

D. $\frac{4}{12}$

True or false?

$$\frac{30}{100} = \frac{3}{100}$$

A. True

B. False

Write two fractions that are equivalent to $\frac{1}{4}$.

Fractions: _____

SET 28 • 4.NF.A.1

BACKWARD 1

Generate simple equivalent fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.b

Find the missing number.

$$\frac{1}{3} = \frac{?}{6}$$

Missing number: _____

True or false?

$$\frac{3}{6} = \frac{1}{2}$$

A. True

B. False

Which fraction is equal to $\frac{1}{4}$?

A. $\frac{4}{1}$

B. $\frac{2}{4}$

C. $\frac{1}{8}$

D. $\frac{2}{8}$

Find the missing number.

$$\frac{2}{4} = \frac{?}{8}$$

Missing number: _____

True or false?

$$\frac{3}{8} = \frac{3}{4}$$

A. True

B. False

SET 28 • 4.NF.A.1

BACKWARD 2

Whole numbers written as fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.c

Write 3 as a fraction with a denominator of 1.

Fraction: _____

True or false?

$$\frac{4}{4} = 1$$

A. True

B. False

Which fraction is equal to 2?

A. $\frac{2}{2}$

B. $\frac{3}{6}$

C. $\frac{6}{3}$

D. $\frac{2}{3}$

Write 1 as a fraction with a denominator of 8.

Fraction: _____

True or false?

$$\frac{6}{6} = 6$$

A. True

B. False

SET 28 • 4.NF.A.1

FORWARD 1

Explain equivalent fractions as multiplying by a fraction equal to 1 • Grade 5 •
Number & Operations–Fractions • 5.NF.B.5.b

True or false?

$\frac{3}{4} \times \frac{2}{2} = \frac{6}{8}$, and $\frac{6}{8} = \frac{3}{4}$ because $\frac{2}{2} = 1$.

A. True

B. False

Which fraction equal to 1 turns $\frac{2}{5}$ into $\frac{6}{15}$?

A. $\frac{3}{3}$

B. $\frac{5}{5}$

C. $\frac{15}{15}$

D. $\frac{6}{6}$

What fraction equal to 1 can you multiply $\frac{5}{8}$ by to get $\frac{20}{32}$?

Fraction: _____

True or false?

$\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$ shows that $\frac{1}{2} = \frac{3}{8}$.

A. True

B. False

Explain why multiplying a fraction by $\frac{5}{5}$ does not change its value.

Explanation: _____

SET 28 • 4.NF.A.1

FORWARD 2

Write fractions as percents • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

Write $\frac{3}{4}$ as a percent.

Percent: _____

True or false?

$$\frac{2}{5} = 40\%$$

A. True

B. False

Which percent is equal to $\frac{7}{10}$?

- A.** 0.7%
- B.** 17%
- C.** 70%
- D.** 7%

Write 35% as a fraction with a denominator of 100.

Fraction: _____

True or false?

$$\frac{1}{8} = 18\%$$

A. True

B. False

Compare two fractions by finding a common denominator or a common numerator

MAIN	Grade 4 main question	4.NF.A.2
BACKWARD 1	Compare fractions with the same numerator or the same denomin...	3.NF.A.3.d
BACKWARD 2	Fractions on a number line	3.NF.A.2
FORWARD 1	Compare the size of a product to the size of one factor	5.NF.B.5.a
FORWARD 2	Compare and order rational numbers	6.NS.C.7.a

SET 29 • 4.NF.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Write $>$, $=$, or $<$.

$$\frac{3}{4} \text{ — } \frac{5}{8}$$

Symbol: _____

Write $>$, $=$, or $<$.

$$\frac{2}{5} \text{ — } \frac{3}{10}$$

Symbol: _____

Which comparison is true?

A. $\frac{3}{8} > \frac{3}{5}$

B. $\frac{3}{5} > \frac{3}{8}$

C. $\frac{3}{5} < \frac{3}{8}$

D. $\frac{3}{5} = \frac{3}{8}$

Compare $\frac{5}{6}$ and $\frac{7}{12}$. Write $>$, $=$, or $<$, and explain how you know.

Comparison: _____

Explanation: _____

True or false?

$$\frac{4}{10} < \frac{35}{100}$$

A. True

B. False

SET 29 • 4.NF.A.2

BACKWARD 1

Compare fractions with the same numerator or the same denominator • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

Write $>$, $=$, or $<$.

$$\frac{3}{8} \text{ — } \frac{5}{8}$$

Symbol: _____

True or false?

$$\frac{1}{3} > \frac{1}{6}$$

A. True

B. False

Which fraction is greatest?

A. $\frac{2}{6}$

B. $\frac{2}{3}$

C. $\frac{2}{4}$

D. $\frac{2}{8}$

Write $>$, $=$, or $<$.

$$\frac{4}{6} \text{ — } \frac{4}{8}$$

Symbol: _____

True or false?

$$\frac{3}{4} < \frac{2}{4}$$

A. True

B. False

SET 29 • 4.NF.A.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point A?



A = _____

True or false?

On a number line, $\frac{5}{6}$ is closer to 1 than $\frac{1}{6}$ is.

A. True

B. False

Into how many equal parts is the space from 0 to 1 divided?



A. 4

B. 7

C. 8

D. 9

What fraction is at point B?



B = _____

True or false?

$\frac{2}{2}$ is at the same point as 2 on a number line.

A. True

B. False

SET 29 • 4.NF.A.2

FORWARD 1

Compare the size of a product to the size of one factor • Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a

True or false?

$\frac{3}{4} \times 9$ is less than 9.

A. True

B. False

Which product is greater than 12?

A. $\frac{5}{4} \times 12$

B. $\frac{3}{4} \times 12$

C. $\frac{1}{2} \times 12$

D. 1×12

Without multiplying, is $\frac{7}{8} \times 20$ greater than, less than, or equal to 20?

Answer: _____

True or false?

$\frac{6}{6} \times 15$ is greater than 15.

A. True

B. False

Without multiplying, which is greater: $\frac{2}{3} \times 30$ or $\frac{4}{3} \times 30$?

Answer: _____

SET 29 • 4.NF.A.2

FORWARD 2

Compare and order rational numbers • Grade 6 • The Number System •
6.NS.C.7.a

Write $>$, $=$, or $<$.

$$-\frac{1}{2} \text{ — } -\frac{3}{4}$$

Symbol: _____

True or false?

$$-\frac{2}{3} < \frac{1}{3}$$

A. True

B. False

Which number is least?

A. $-\frac{5}{6}$

B. 0

C. $\frac{1}{4}$

D. $-\frac{1}{2}$

Order from least to greatest.

$$\frac{1}{2}, -\frac{3}{4}, -\frac{1}{4}$$

Order: _____

True or false?

$$-\frac{1}{4} < -\frac{1}{2}$$

A. True

B. False

Compare fractions using the benchmark $\frac{1}{2}$, and only when they refer to the same whole

MAIN	Grade 4 main question	4.NF.A.2
BACKWARD 1	Compare fractions of the same whole	3.NF.A.3.d
BACKWARD 2	Equal parts of a shape	3.G.A.2
FORWARD 1	Use benchmark fractions to estimate and check sums and differe...	5.NF.A.2
FORWARD 2	Write and interpret statements of order for rational numbers in c...	6.NS.C.7.b

SET 30 • 4.NF.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which fraction is greater than $\frac{1}{2}$?

A. $\frac{2}{5}$

B. $\frac{3}{8}$

C. $\frac{5}{8}$

D. $\frac{4}{10}$

Compare $\frac{3}{8}$ and $\frac{4}{6}$ by comparing each one to $\frac{1}{2}$. Write $>$, $=$, or $<$.

Comparison: _____

Reason: _____

True or false?

$$\frac{5}{12} < \frac{1}{2}$$

A. True

B. False

True or false?

Ben ate $\frac{1}{2}$ of a small pizza and Kim ate $\frac{1}{2}$ of a large pizza. They ate the same amount of pizza.

- A.** True
- B.** False

Explain why $\frac{2}{3}$ of a small cake and $\frac{2}{3}$ of a large cake are not the same amount of cake.

Explanation: _____

SET 30 • 4.NF.A.2

BACKWARD 1

Compare fractions of the same whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

True or false?

$\frac{1}{4}$ of a pizza is less than $\frac{1}{2}$ of the same pizza.

A. True

B. False

Write $>$, $=$, or $<$.

$$\frac{2}{6} \text{ — } \frac{5}{6}$$

Symbol: _____

Which fraction of the same bar is greatest?

A. $\frac{1}{8}$

B. $\frac{1}{4}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

Write $>$, $=$, or $<$.

$$\frac{3}{4} \text{ — } \frac{3}{6}$$

Symbol: _____

True or false?

$\frac{1}{8}$ of a ribbon is more than $\frac{1}{3}$ of the same ribbon.

A. True

B. False

SET 30 • 4.NF.A.2

BACKWARD 2

Equal parts of a shape • Grade 3 • Geometry • 3.G.A.2

A square is split into 4 equal parts.
What fraction of its area is each part?

Fraction: _____

True or false?

A rectangle cut into 6 equal parts has parts that are each $\frac{1}{6}$ of its area.

A. True

B. False

A circle is cut into 8 equal parts.

What fraction of the area is 1 part?

A. $\frac{8}{1}$

B. $\frac{7}{8}$

C. $\frac{1}{4}$

D. $\frac{1}{8}$

What fraction of the shape is shaded?



Fraction: _____

True or false?

A shape cut into 2 parts of different sizes is cut into halves.

A. True

B. False

SET 30 • 4.NF.A.2

FORWARD 1

Use benchmark fractions to estimate and check sums and differences • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

True or false?

$\frac{3}{5} + \frac{4}{7}$ is greater than 1, because each fraction is greater than $\frac{1}{2}$.

A. True

B. False

Which is the best estimate of $\frac{7}{8} + \frac{1}{12}$?

- A.** About 1
- B.** About 0
- C.** About 2
- D.** About $\frac{1}{2}$

Use benchmark fractions to decide whether $\frac{2}{5} + \frac{3}{8}$ is more or less than 1.

Answer: _____

Reason: _____

True or false?

$\frac{5}{6} - \frac{1}{4}$ is about 1, because both fractions are close to 1.

- A.** True
- B.** False

Use the benchmarks 0, $\frac{1}{2}$, and 1 to estimate $\frac{9}{10} + \frac{5}{12}$.

Estimate: _____

SET 30 • 4.NF.A.2

FORWARD 2

Write and interpret statements of order for rational numbers in context • Grade 6 • The Number System • 6.NS.C.7.b

Write an inequality that compares $-\frac{3}{4}^{\circ}\text{C}$ and $-\frac{1}{2}^{\circ}\text{C}$.

Inequality: _____

True or false?

A temperature of $-\frac{1}{4}^{\circ}\text{F}$ is warmer than $-\frac{3}{4}^{\circ}\text{F}$.

A. True

B. False

Which statement is true?

A. $-\frac{2}{3} < -\frac{1}{3}$

B. $-\frac{2}{3} > -\frac{1}{3}$

C. $-\frac{1}{3} < -\frac{2}{3}$

D. $\frac{1}{3} < -\frac{2}{3}$

A diver at $-2\frac{1}{2}$ m is deeper than a diver at $-1\frac{1}{2}$ m.

Write an inequality for the two positions.

Inequality: _____

True or false?

$$-\frac{1}{8} < -\frac{7}{8}$$

A. True

B. False

Understand adding and subtracting fractions as joining and separating parts of the same whole

MAIN	Grade 4 main question	4.NF.B.3.a
BACKWARD 1	A fraction is a number of unit-fraction parts	3.NF.A.1
BACKWARD 2	Add to, take from, put together, and take apart within 20	1.OA.A.1
FORWARD 1	Add and subtract fractions with unlike denominators	5.NF.A.1
FORWARD 2	Solve equations of the form $x + p = q$ with fractions	6.EE.B.7

SET 31 • 4.NF.B.3.a

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.a

A pizza is cut into 8 equal slices. Jo eats 3 slices and Al eats 2 slices.

What fraction of the pizza did they eat in all?

Fraction: _____

True or false?

$\frac{2}{5} + \frac{1}{5}$ means joining 2 fifths and 1 fifth of the same whole, which makes $\frac{3}{5}$.

A. True

B. False

A ribbon is cut into 10 equal pieces. 3 pieces are used and 7 are left.

Which equation shows this?

A. $\frac{10}{10} - \frac{3}{10} = \frac{7}{20}$

B. $\frac{3}{10} - \frac{10}{10} = \frac{7}{10}$

C. $\frac{10}{10} - \frac{3}{10} = \frac{7}{0}$

D. $\frac{10}{10} - \frac{3}{10} = \frac{7}{10}$

True or false?

$\frac{3}{4} + \frac{1}{4} = \frac{4}{8}$, because you add the numerators and add the denominators.

A. True

B. False

Shade the bar to show $\frac{2}{6} + \frac{3}{6}$.

Then write the sum.

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Sum: _____

SET 31 • 4.NF.B.3.a

BACKWARD 1

A fraction is a number of unit-fraction parts • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{3}{8}$ is 3 parts of size $\frac{1}{8}$.

A. True

B. False

How many $\frac{1}{6}$ s make $\frac{5}{6}$?

Answer: _____

What fraction is 2 parts of size $\frac{1}{3}$?

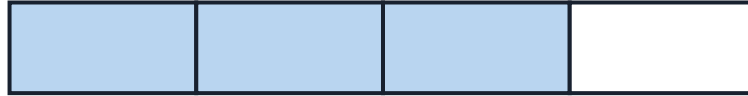
A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. $\frac{3}{2}$

D. $\frac{2}{6}$

What fraction of the bar is shaded?



Fraction: _____

True or false?

$\frac{4}{6}$ means 6 parts of size $\frac{1}{4}$.

A. True

B. False

SET 31 • 4.NF.B.3.a

BACKWARD 2

Add to, take from, put together, and take apart within 20 • Grade 1 •
Operations & Algebraic Thinking • 1.OA.A.1

Sam has 7 marbles. He gets 6 more.

How many marbles does he have?

Marbles: _____

True or false?

There are 15 birds. 8 fly away. 7 birds are left.

A. True

B. False

9 red apples and 5 green apples are put together.

How many apples are there?

A. 14

B. 13

C. 4

D. 45

A box has 12 crayons. 4 are taken out.

How many crayons are left?

Crayons: _____

True or false?

6 cats and 7 dogs make 12 pets.

A. True

B. False

SET 31 • 4.NF.B.3.a

FORWARD 1

Add and subtract fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{1}{4} + \frac{2}{3}$$

Sum: _____

Subtract.

$$\frac{7}{10} - \frac{1}{2}$$

Difference: _____

Add.

$$\frac{3}{8} + \frac{1}{2}$$

A. $\frac{4}{10}$

B. $\frac{4}{8}$

C. $\frac{7}{8}$

D. $1\frac{1}{8}$

True or false?

$$\frac{5}{6} - \frac{1}{3} = \frac{1}{2}$$

A. True

B. False

True or false?

$$\frac{1}{5} + \frac{1}{2} = \frac{2}{7}$$

A. True

B. False

SET 31 • 4.NF.B.3.a

FORWARD 2

Solve equations of the form $x + p = q$ with fractions • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + \frac{1}{4} = \frac{3}{4}$$

x = _____

True or false?

The solution of $y - \frac{2}{5} = \frac{1}{5}$ is $y = \frac{3}{5}$.

A. True

B. False

Solve for n.

$$n + \frac{3}{8} = 1$$

A. $1\frac{3}{8}$

B. $\frac{3}{8}$

C. $\frac{5}{8}$

D. $\frac{8}{5}$

Solve for m.

$$\frac{5}{6} = m + \frac{1}{6}$$

m = _____

True or false?

The solution of $x + \frac{1}{3} = \frac{2}{3}$ is $x = 1$.

A. True

B. False

Decompose a fraction into a sum of fractions with the same denominator, and justify it with a model

MAIN	Grade 4 main question	4.NF.B.3.b
BACKWARD 1	Unit fractions build a fraction	3.NF.A.1
BACKWARD 2	Break apart numbers to 10 into pairs in more than one way	K.OA.A.3
FORWARD 1	Add and subtract mixed numbers with unlike denominators	5.NF.A.1
FORWARD 2	Divide fractions by thinking "how many parts of this size?"	6.NS.A.1

SET 32 • 4.NF.B.3.b

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.b

Write $\frac{4}{5}$ as a sum of unit fractions.

Sum: _____

Write $\frac{7}{10}$ as a sum of two fractions with a denominator of 10, in two different ways.

Way 1: _____

Way 2: _____

Which sum is equal to $\frac{5}{6}$?

A. $\frac{1}{6} + \frac{5}{6}$

B. $\frac{2}{6} + \frac{3}{6}$

C. $\frac{2}{3} + \frac{3}{3}$

D. $\frac{4}{6} + \frac{2}{6}$

True or false?

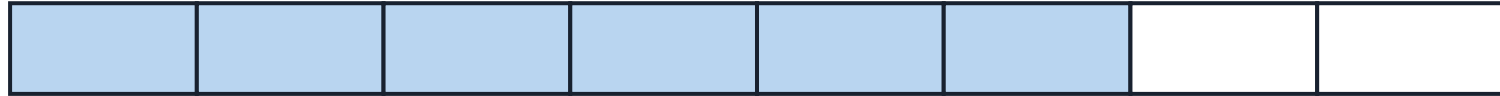
$$1\frac{2}{3} = \frac{3}{3} + \frac{2}{3}$$

A. True

B. False

The bar shows $\frac{6}{8}$ shaded.

Circle a group of 2 eighths and a group of 4 eighths to show $\frac{6}{8} = \frac{2}{8} + \frac{4}{8}$. Then write another way to decompose $\frac{6}{8}$.



Another way: _____

SET 32 • 4.NF.B.3.b

BACKWARD 1

Unit fractions build a fraction • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{8}$ s make $\frac{7}{8}$?

Answer: _____

True or false?

$$\frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$$

A. True

B. False

What fraction is 5 parts of size $\frac{1}{6}$?

A. $\frac{5}{1}$

B. $\frac{5}{6}$

C. $\frac{1}{30}$

D. $\frac{6}{5}$

Write $\frac{2}{3}$ as a sum of unit fractions.

Sum: _____

True or false?

$$\frac{1}{2} + \frac{1}{2} = \frac{2}{4}$$

A. True

B. False

SET 32 • 4.NF.B.3.b

BACKWARD 2

Break apart numbers to 10 into pairs in more than one way • Grade K •
Operations & Algebraic Thinking • K.OA.A.3

Write 7 as the sum of two numbers in two different ways.

Way 1: _____

Way 2: _____

True or false?

$$6 = 2 + 4$$

A. True

B. False

Which pair of numbers makes 9?

A. 5 and 3

B. 5 and 4

C. 7 and 3

D. 6 and 4

Fill in the blank.

$$8 = 5 + \underline{\quad}$$

Blank: _____

True or false?

$$10 = 6 + 3$$

A. True

B. False

SET 32 • 4.NF.B.3.b

FORWARD 1

Add and subtract mixed numbers with unlike denominators • Grade 5 •
Number & Operations–Fractions • 5.NF.A.1

Add.

$$1\frac{1}{2} + 2\frac{1}{3}$$

Sum: _____

Subtract.

$$3\frac{3}{4} - 1\frac{1}{6}$$

Difference: _____

Add.

$$2\frac{2}{5} + 1\frac{1}{2}$$

A. $3\frac{3}{7}$

B. $4\frac{1}{10}$

C. $3\frac{9}{10}$

D. $3\frac{3}{10}$

True or false?

$$4\frac{1}{2} - 1\frac{3}{4} = 2\frac{3}{4}$$

A. True

B. False

True or false?

$$1\frac{2}{3} + 1\frac{1}{6} = 2\frac{3}{9}$$

A. True

B. False

SET 32 • 4.NF.B.3.b

FORWARD 2

Divide fractions by thinking "how many parts of this size?" • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{1}{8}$ s are in $\frac{3}{4}$?

Answer: _____

True or false?

$$\frac{2}{3} \div \frac{1}{6} = 4$$

A. True

B. False

Divide.

$$\frac{5}{6} \div \frac{5}{12}$$

A. 10

B. $\frac{25}{72}$

C. $\frac{1}{2}$

D. 2

Divide.

$$\frac{9}{10} \div \frac{3}{10}$$

Quotient: _____

True or false?

$$\frac{1}{2} \div \frac{1}{4} = \frac{1}{8}$$

A. True

B. False

Add mixed numbers with like denominators

MAIN

Grade 4 main question

4.NF.B.3.c

BACKWARD 1

Whole numbers written as fractions

3.NF.A.3.c

BACKWARD 2

Fractions greater than 1 on a number line

3.NF.A.2

FORWARD 1

Add mixed numbers with unlike denominators

5.NF.A.1

FORWARD 2

Solve equations of the form $x + p = q$ with mixed numbers

6.EE.B.7

SET 33 • 4.NF.B.3.c

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.c

Add.

$$2\frac{3}{8} + 1\frac{4}{8}$$

Sum: _____

Add.

$$3\frac{4}{5} + 2\frac{3}{5}$$

Sum: _____

Add.

$$1\frac{5}{6} + 4\frac{2}{6}$$

A. $6\frac{1}{6}$

B. $5\frac{7}{12}$

C. $6\frac{7}{12}$

D. $5\frac{1}{6}$

True or false?

$$2\frac{1}{4} + 1\frac{3}{4} = 3\frac{4}{8}$$

A. True

B. False

A plant grew $1\frac{7}{10}$ inches in May and $2\frac{6}{10}$ inches in June.

How much did it grow in all?

Growth: _____

SET 33 • 4.NF.B.3.c

BACKWARD 1

Whole numbers written as fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.c

Write 2 as a fraction with a denominator of 4.

Fraction: _____

True or false?

$$\frac{3}{3} = 1$$

A. True

B. False

Which fraction is equal to 1?

A. $\frac{0}{6}$

B. $\frac{6}{6}$

C. $\frac{6}{1}$

D. $\frac{1}{6}$

How many halves are in 3 wholes?

Halves: _____

True or false?

$$\frac{8}{4} = 4$$

A. True

B. False

SET 33 • 4.NF.B.3.c

BACKWARD 2

Fractions greater than 1 on a number line • Grade 3 • Number & Operations–Fractions • 3.NF.A.2

What number is at point A? Write a fraction.



A = _____

True or false?

$\frac{3}{2}$ is between 1 and 2 on a number line.

A. True

B. False

Which fraction is at the same point as 1 on a number line?

A. $\frac{1}{4}$

B. $\frac{4}{1}$

C. $\frac{3}{4}$

D. $\frac{4}{4}$

What number is at point B? Write a fraction.



B = _____

True or false?

$\frac{7}{8}$ is greater than 1.

A. True

B. False

SET 33 • 4.NF.B.3.c

FORWARD 1

Add mixed numbers with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$3\frac{1}{4} + 1\frac{2}{3}$$

Sum: _____

Add.

$$2\frac{5}{6} + 1\frac{1}{2}$$

Sum: _____

Add.

$$1\frac{3}{10} + 2\frac{1}{2}$$

A. $3\frac{4}{5}$

B. $4\frac{4}{5}$

C. $3\frac{2}{5}$

D. $3\frac{4}{12}$

True or false?

$$4\frac{3}{8} + 2\frac{1}{4} = 6\frac{5}{8}$$

A. True

B. False

True or false?

$$1\frac{1}{2} + 1\frac{1}{3} = 2\frac{2}{5}$$

A. True

B. False

SET 33 • 4.NF.B.3.c

FORWARD 2

Solve equations of the form $x + p = q$ with mixed numbers • Grade 6 •
Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 1\frac{1}{2} = 4\frac{1}{4}$$

x = _____

True or false?

The solution of $y - 2\frac{1}{3} = 1\frac{2}{3}$ is $y = 4$.

A. True

B. False

Solve for n.

$$n + 3\frac{3}{8} = 6$$

A. $2\frac{3}{8}$

B. $2\frac{5}{8}$

C. $3\frac{5}{8}$

D. $9\frac{3}{8}$

A board was $6\frac{1}{2}$ feet long. After a piece was cut off, $4\frac{3}{4}$ feet were left.

Write and solve an equation for the length p of the piece.

Equation: _____

$p =$ _____

True or false?

The solution of $x + 2\frac{1}{2} = 5$ is $x = 3\frac{1}{2}$.

A. True

B. False

Subtract mixed numbers with like denominators

MAIN

Grade 4 main question

4.NF.B.3.c

BACKWARD 1

Whole numbers written as fractions

3.NF.A.3.c

BACKWARD 2

Subtract within 100

2.NBT.B.5

FORWARD 1

Subtract mixed numbers with unlike denominators

5.NF.A.1

FORWARD 2

Solve equations with subtraction and mixed numbers

6.EE.B.7

SET 34 • 4.NF.B.3.c

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.c

Subtract.

$$5\frac{7}{8} - 2\frac{3}{8}$$

Difference: _____

Subtract.

$$4\frac{1}{5} - 1\frac{3}{5}$$

Difference: _____

Subtract.

$$6 - 2\frac{5}{6}$$

A. $3\frac{5}{6}$

B. $4\frac{5}{6}$

C. $4\frac{1}{6}$

D. $3\frac{1}{6}$

True or false?

$$3\frac{2}{10} - 1\frac{7}{10} = 1\frac{5}{10}$$

A. True

B. False

A jug had $3\frac{1}{4}$ liters of water. Lee poured out $1\frac{3}{4}$ liters.

How much water is left?

Water left: _____

SET 34 • 4.NF.B.3.c

BACKWARD 1

Whole numbers written as fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.c

Write 1 as a fraction with a denominator of 6.

Fraction: _____

True or false?

$$\frac{4}{4} = 1$$

A. True

B. False

Which fraction is equal to 3?

A. $\frac{4}{12}$

B. $\frac{3}{4}$

C. $\frac{4}{4}$

D. $\frac{12}{4}$

How many thirds are in 2 wholes?

Thirds: _____

True or false?

$$\frac{2}{2} = 2$$

A. True

B. False

SET 34 • 4.NF.B.3.c

BACKWARD 2

Subtract within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Subtract.

$$62 - 37$$

Difference: _____

True or false?

$$80 - 26 = 54$$

A. True

B. False

$$71 - 48 = ?$$

A. 119

B. 23

C. 33

D. 27

Subtract.

$$95 - 59$$

Difference: _____

True or false?

$$43 - 18 = 35$$

A. True

B. False

SET 34 • 4.NF.B.3.c

FORWARD 1

Subtract mixed numbers with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$5\frac{1}{2} - 2\frac{1}{3}$$

Difference: _____

Subtract.

$$4 - 1\frac{3}{8}$$

Difference: _____

Subtract.

$$3\frac{1}{4} - 1\frac{2}{3}$$

A. $2\frac{1}{12}$

B. $1\frac{7}{12}$

C. $1\frac{5}{12}$

D. $2\frac{7}{12}$

True or false?

$$6\frac{1}{2} - 3\frac{3}{4} = 2\frac{3}{4}$$

A. True

B. False

True or false?

$$2\frac{5}{6} - 1\frac{1}{2} = 1\frac{4}{4}$$

A. True

B. False

SET 34 • 4.NF.B.3.c

FORWARD 2

Solve equations with subtraction and mixed numbers • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x - 2\frac{1}{4} = 3\frac{1}{2}$$

x = _____

True or false?

The solution of $y + 1\frac{5}{6} = 4$ is $y = 2\frac{1}{6}$.

A. True

B. False

Solve for n.

$$7\frac{1}{3} - n = 2\frac{2}{3}$$

A. $4\frac{2}{3}$

B. 10

C. $4\frac{1}{3}$

D. $5\frac{1}{3}$

A tank had some water. After $3\frac{1}{2}$ gallons were used, $5\frac{1}{4}$ gallons were left.

Write and solve an equation for the starting amount w .

Equation: _____

$w =$ _____

True or false?

The solution of $x - 1\frac{1}{2} = 1\frac{1}{2}$ is $x = 0$.

A. True

B. False

Solve word problems by adding and subtracting fractions with like denominators

MAIN	Grade 4 main question	4.NF.B.3.d
BACKWARD 1	One- and two-step word problems within 100	2.OA.A.1
BACKWARD 2	Fractions of a whole	3.NF.A.1
FORWARD 1	Solve word problems by adding and subtracting fractions with unl...	5.NF.A.2
FORWARD 2	Solve word problems by dividing fractions	6.NS.A.1

SET 35 • 4.NF.B.3.d

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.d

Mia ran $\frac{3}{10}$ mile in the morning and $\frac{5}{10}$ mile in the afternoon.

How far did she run in all?

Distance: _____

A jug had $\frac{11}{12}$ liter of juice. Tom drank $\frac{4}{12}$ liter.

How much juice is left?

Juice left: _____

Ty used $1\frac{2}{4}$ cups of rice on Monday and $2\frac{3}{4}$ cups on Tuesday.

How many cups did he use in all?

A. $3\frac{1}{4}$ cups

B. $3\frac{5}{8}$ cups

C. $4\frac{1}{4}$ cups

D. $4\frac{5}{8}$ cups

True or false?

A ribbon is $\frac{7}{8}$ yard long. Ann cuts off $\frac{3}{8}$ yard. $\frac{4}{8}$ yard is left.

- A. True
- B. False

A painter used $\frac{2}{6}$ of a can of paint on a door and $\frac{3}{6}$ of the can on a fence.

What fraction of the can is left?

Fraction: _____

SET 35 • 4.NF.B.3.d

BACKWARD 1

One- and two-step word problems within 100 • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

A shelf has 46 books. 18 are taken.

How many books are left?

Books: _____

True or false?

Kim had 25 stickers and got 17 more. She has 42 stickers.

- A.** True
- B.** False

A class has 14 boys and 15 girls.

How many students are in the class?

A. 39

B. 19

C. 1

D. 29

Jo had 50 cents. She spent 23 cents.

How much money is left?

Money left: _____

True or false?

A bus had 36 people. 9 got off. 25 people are still on the bus.

A. True

B. False

SET 35 • 4.NF.B.3.d

BACKWARD 2

Fractions of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

A pie is cut into 6 equal pieces. 5 pieces are eaten.
What fraction of the pie is eaten?

Fraction: _____

True or false?

A pizza is cut into 8 equal slices and 3 are left. $\frac{3}{8}$ of the pizza is left.

- A. True
- B. False

A bar is cut into 4 equal parts.

What fraction of the bar is 1 part?

A. $\frac{3}{4}$

B. $\frac{1}{3}$

C. $\frac{1}{4}$

D. $\frac{4}{1}$

A garden has 3 equal sections. Beans fill 2 of them.
What fraction of the garden is beans?

Fraction: _____

True or false?

A board cut into 2 equal pieces has pieces that are each $\frac{1}{3}$ of the board.

- A. True
- B. False

SET 35 • 4.NF.B.3.d

FORWARD 1

Solve word problems by adding and subtracting fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

Ana walked $\frac{1}{2}$ mile and then $\frac{1}{3}$ mile.

How far did she walk in all?

Distance: _____

A jug had $\frac{3}{4}$ gallon of milk. Ben used $\frac{2}{3}$ gallon.

How much milk is left?

Milk left: _____

Lu read $\frac{1}{4}$ of a book on Monday and $\frac{3}{8}$ of it on Tuesday.

What fraction of the book did she read?

A. $\frac{4}{12}$

B. $\frac{5}{8}$

C. $\frac{4}{8}$

D. $\frac{3}{32}$

True or false?

A board is $\frac{5}{6}$ meter long. $\frac{1}{2}$ meter is cut off. $\frac{1}{3}$ meter is left.

A. True

B. False

True or false?

Sam ate $\frac{1}{2}$ of a pizza and Lee ate $\frac{1}{3}$ of it. They ate $\frac{2}{5}$ of the pizza.

A. True

B. False

SET 35 • 4.NF.B.3.d

FORWARD 2

Solve word problems by dividing fractions • Grade 6 • The Number System •
6.NS.A.1

How many $\frac{1}{4}$ -cup servings are in $\frac{3}{4}$ cup?

Servings: _____

True or false?

A $\frac{9}{10}$ -mile path is split into $\frac{3}{10}$ -mile sections. There are 3 sections.

A. True

B. False

How many $\frac{2}{3}$ -foot pieces can be cut from 4 feet of ribbon?

A. 6

B. 8

C. $\frac{1}{6}$

D. $\frac{8}{3}$

$\frac{3}{4}$ pound of nuts is shared equally among 3 bags.

How much is in each bag?

Pounds: _____

True or false?

$\frac{1}{2}$ cup of oats fills a $\frac{1}{6}$ -cup scoop exactly 2 times.

A. True

B. False

Understand a fraction a/b as a multiple of $1/b$

MAIN	Grade 4 main question	4.NF.B.4.a
BACKWARD 1	A fraction is a number of unit-fraction parts	3.NF.A.1
BACKWARD 2	Products as equal groups	3.OA.A.1
FORWARD 1	Multiply a fraction by a whole number as a part of a whole number	5.NF.B.4.a
FORWARD 2	Divide a fraction by a unit fraction	6.NS.A.1

SET 36 • 4.NF.B.4.a

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.4.a

Write $\frac{7}{8}$ as a whole number times $\frac{1}{8}$.

Expression: _____

Which equation shows $\frac{3}{5}$ as a multiple of $\frac{1}{5}$?

A. $\frac{3}{5} = \frac{1}{5} \div 3$

B. $\frac{3}{5} = 3 + \frac{1}{5}$

C. $\frac{3}{5} = 5 \times \frac{1}{3}$

D. $\frac{3}{5} = 3 \times \frac{1}{5}$

True or false?

$$\frac{9}{4} = 9 \times \frac{1}{4}$$

A. True

B. False

Shade the tape diagram to show $4 \times \frac{1}{6}$.

Then write the product as a fraction.

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Fraction: _____

True or false?

$$\frac{6}{10} = 10 \times \frac{1}{6}$$

A. True

B. False

SET 36 • 4.NF.B.4.a

BACKWARD 1

A fraction is a number of unit-fraction parts • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

How many $\frac{1}{4}$ s make $\frac{3}{4}$?

Answer: _____

True or false?

$\frac{5}{8}$ is 5 parts of size $\frac{1}{8}$.

A. True

B. False

What fraction is 4 parts of size $\frac{1}{6}$?

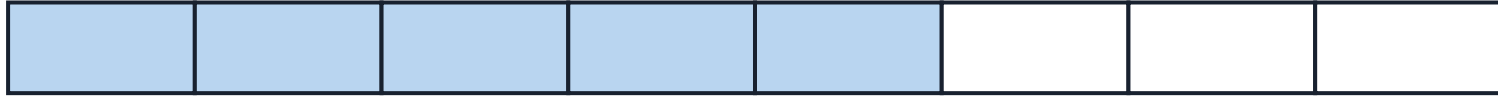
A. $\frac{4}{24}$

B. $\frac{1}{24}$

C. $\frac{4}{6}$

D. $\frac{6}{4}$

What fraction of the bar is shaded?



Fraction: _____

True or false?

$\frac{2}{3}$ is 3 parts of size $\frac{1}{2}$.

A. True

B. False

SET 36 • 4.NF.B.4.a

BACKWARD 2

Products as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

What is the total in 5 groups of 3?

Total: _____

True or false?

4×6 can mean 4 groups of 6.

A. True

B. False

Which expression shows 7 groups of 2?

A. 7×2

B. $2 \div 7$

C. $7 - 2$

D. $7 + 2$

Write a multiplication expression for 3 groups of 9.

Expression: _____

True or false?

6×3 means $6 + 3$.

A. True

B. False

SET 36 • 4.NF.B.4.a

FORWARD 1

Multiply a fraction by a whole number as a part of a whole number • Grade 5 •
Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{2}{3} \times 6$$

Product: _____

True or false?

$$\frac{3}{4} \times 8 = 6$$

A. True

B. False

What is $\frac{2}{5}$ of 15?

A. 10

B. 30

C. 3

D. 6

Multiply.

$$\frac{5}{6} \times 4$$

Product: _____

True or false?

$$\frac{1}{2} \times 10 = 20$$

A. True

B. False

SET 36 • 4.NF.B.4.a

FORWARD 2

Divide a fraction by a unit fraction • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{1}{5}$ s are in $\frac{4}{5}$?

Answer: _____

True or false?

$$\frac{7}{8} \div \frac{1}{8} = 7$$

A. True

B. False

Divide.

$$\frac{3}{4} \div \frac{1}{8}$$

A. $\frac{3}{32}$

B. 6

C. 3

D. $\frac{1}{6}$

Divide.

$$2 \div \frac{1}{3}$$

Quotient: _____

True or false?

$$\frac{2}{3} \div \frac{1}{3} = \frac{2}{9}$$

A. True

B. False

Multiply a fraction by a whole number:

$$n \times (a/b) = (n \times a)/b$$

MAIN

Grade 4 main question

4.NF.B.4.b

BACKWARD 1

Multiply three numbers in any grouping

3.OA.B.5

BACKWARD 2

A fraction is a number of unit-fraction parts

3.NF.A.1

FORWARD 1

Multiply a fraction by a fraction

5.NF.B.4.a

FORWARD 2

Divide a fraction by a fraction

6.NS.A.1

SET 37 • 4.NF.B.4.b

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.4.b

Multiply.

$$4 \times \frac{2}{3}$$

Product: _____

Multiply.

$$5 \times \frac{3}{8}$$

Product: _____

Which is equal to $3 \times \frac{4}{5}$?

A. $12 \times \frac{1}{5}$

B. $7 \times \frac{1}{5}$

C. $3 \times \frac{1}{5}$

D. $\frac{12}{15}$

True or false?

$$6 \times \frac{2}{12} = \frac{12}{72}$$

A. True

B. False

Explain why $2 \times \frac{3}{10} = 6 \times \frac{1}{10}$.

Explanation: _____

SET 37 • 4.NF.B.4.b

BACKWARD 1

Multiply three numbers in any grouping • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 5 = 2 \times (3 \times 5)$$

A. True

B. False

Fill in the blank.

$$4 \times (2 \times 3) = (4 \times 2) \times \underline{\hspace{1cm}}$$

Blank: _____

Which is equal to $3 \times 2 \times 4$?

A. 6×4

B. 5×4

C. 9×4

D. $3 + 8$

Multiply.

$$2 \times 4 \times 5$$

Product: _____

True or false?

$$(3 \times 2) \times 4 = 3 + 2 + 4$$

A. True

B. False

SET 37 • 4.NF.B.4.b

BACKWARD 2

A fraction is a number of unit-fraction parts • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

How many $\frac{1}{3}$ s make $\frac{2}{3}$?

Answer: _____

True or false?

$\frac{6}{8}$ is 6 parts of size $\frac{1}{8}$.

A. True

B. False

What fraction is 3 parts of size $\frac{1}{4}$?

A. $\frac{1}{12}$

B. $\frac{3}{4}$

C. $\frac{3}{12}$

D. $\frac{4}{3}$

What fraction is 5 parts of size $\frac{1}{6}$?

Fraction: _____

True or false?

$\frac{3}{6}$ is 6 parts of size $\frac{1}{3}$.

A. True

B. False

SET 37 • 4.NF.B.4.b

FORWARD 1

Multiply a fraction by a fraction • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{2}{3} \times \frac{3}{5}$$

Product: _____

True or false?

$$\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$$

A. True

B. False

Multiply.

$$\frac{3}{4} \times \frac{2}{5}$$

A. $\frac{3}{10}$

B. $\frac{5}{9}$

C. $\frac{5}{20}$

D. $\frac{6}{9}$

What is $\frac{1}{3}$ of $\frac{6}{7}$?

Answer: _____

True or false?

$$\frac{2}{5} \times \frac{1}{2} = \frac{3}{7}$$

A. True

B. False

SET 37 • 4.NF.B.4.b

FORWARD 2

Divide a fraction by a fraction • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{2}{3} \div \frac{1}{6}$$

Quotient: _____

True or false?

$$\frac{3}{4} \div \frac{3}{8} = 2$$

A. True

B. False

Divide.

$$\frac{4}{5} \div \frac{2}{5}$$

A. 2

B. $\frac{1}{2}$

C. 4

D. $\frac{8}{25}$

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

Quotient: _____

True or false?

$$\frac{1}{2} \div \frac{1}{8} = \frac{1}{16}$$

A. True

B. False

Solve word problems by multiplying a fraction by a whole number

MAIN	Grade 4 main question	4.NF.B.4.c
BACKWARD 1	Equal-groups word problems within 100	3.OA.A.3
BACKWARD 2	Jump by a unit fraction on a number line	3.NF.A.2
FORWARD 1	Solve real-world problems by multiplying fractions and mixed num...	5.NF.B.6
FORWARD 2	Find a percent of a quantity	6.RP.A.3.c

SET 38 • 4.NF.B.4.c

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.4.c

Each glass holds $\frac{3}{4}$ cup of milk.

How much milk is in 6 glasses?

Milk: _____

A snail crawls $\frac{2}{5}$ meter each hour.

How far does it crawl in 4 hours?

Distance: _____

Each of 3 friends eats $\frac{5}{8}$ of a sandwich.

How many sandwiches do they eat in all?

A. $1\frac{7}{8}$

B. $\frac{8}{8}$

C. $\frac{15}{24}$

D. $3\frac{5}{8}$

True or false?

A runner runs $\frac{7}{10}$ mile each day for 5 days. She runs $3\frac{5}{10}$ miles in all.

A. True

B. False

A recipe uses $\frac{2}{3}$ cup of flour for each batch.

How much flour do 5 batches use, and between which two whole numbers is that amount?

Flour: _____

Between: _____

SET 38 • 4.NF.B.4.c

BACKWARD 1

Equal-groups word problems within 100 • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

Each bag holds 6 oranges.

How many oranges are in 7 bags?

Oranges: _____

True or false?

9 boxes with 4 pens in each box hold 36 pens.

A. True

B. False

A spider has 8 legs.

How many legs do 5 spiders have?

A. 35

B. 45

C. 13

D. 40

Each row has 9 seats.

How many seats are in 3 rows?

Seats: _____

True or false?

6 packs of 5 cards is 11 cards.

A. True

B. False

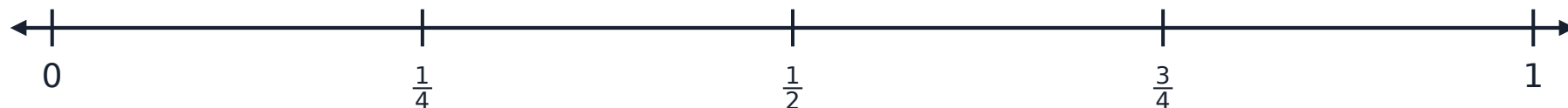
SET 38 • 4.NF.B.4.c

BACKWARD 2

Jump by a unit fraction on a number line • Grade 3 • Number & Operations–Fractions • 3.NF.A.2

Start at 0 and make 3 jumps of $\frac{1}{4}$ on the number line.

Where do you land?



Answer: _____

True or false?

5 jumps of $\frac{1}{8}$ from 0 land on $\frac{5}{8}$.

A. True

B. False

How many jumps of $\frac{1}{3}$ from 0 reach 1?

A. 3

B. 2

C. 6

D. 1

Where do 4 jumps of $\frac{1}{6}$ from 0 land?

Answer: _____

True or false?

2 jumps of $\frac{1}{2}$ from 0 land on $\frac{2}{4}$.

A. True

B. False

SET 38 • 4.NF.B.4.c

FORWARD 1

Solve real-world problems by multiplying fractions and mixed numbers • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A recipe uses $\frac{3}{4}$ cup of oats per batch.

How many cups of oats are in $2\frac{1}{2}$ batches?

Cups: _____

A garden is 10 m long. $\frac{3}{5}$ of its length is planted.

How many meters are planted?

Meters: _____

A bottle holds $1\frac{1}{2}$ liters. It is $\frac{2}{3}$ full.

How many liters are in the bottle?

A. $\frac{1}{2}$ liter

B. $2\frac{1}{6}$ liters

C. 1 liter

D. $\frac{3}{4}$ liter

True or false?

A board is $4\frac{1}{2}$ feet long. Half of the board is $2\frac{1}{4}$ feet long.

- A.** True
- B.** False

True or false?

$\frac{2}{3}$ of a 9-mile trail is 3 miles.

A. True

B. False

SET 38 • 4.NF.B.4.c

FORWARD 2

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

What is 25% of 36?

Answer: _____

True or false?

50% of 18 is 9.

A. True

B. False

What is 75% of 40?

A. 10

B. 35

C. 25

D. 30

A class has 20 students. 40% of them walk to school.
How many students walk?

Students: _____

True or false?

10% of 70 is 10.

A. True

B. False

Write tenths as hundredths and add fractions with denominators 10 and 100

MAIN	Grade 4 main question	4.NF.C.5
BACKWARD 1	Simple equivalent fractions	3.NF.A.3.b
BACKWARD 2	A hundred is ten tens	2.NBT.A.1.a
FORWARD 1	Add decimals to hundredths	5.NBT.B.7
FORWARD 2	Write tenths and hundredths as percents	6.RP.A.3.c

SET 39 • 4.NF.C.5

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

Write $\frac{6}{10}$ as an equivalent fraction with a denominator of 100.

Fraction: _____

Add.

$$\frac{2}{10} + \frac{45}{100}$$

Sum: _____

$$\frac{7}{10} + \frac{9}{100} = ?$$

A. $\frac{79}{100}$

B. $\frac{16}{110}$

C. $\frac{16}{100}$

D. $\frac{79}{10}$

True or false?

$$\frac{50}{100} = \frac{5}{100}$$

A. True

B. False

Add.

$$\frac{8}{10} + \frac{15}{100}$$

Sum: _____

SET 39 • 4.NF.C.5

BACKWARD 1

Simple equivalent fractions • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.b

Find the missing number.

$$\frac{1}{2} = \frac{?}{8}$$

Missing number: _____

True or false?

$$\frac{2}{4} = \frac{4}{8}$$

A. True

B. False

Which fraction is equal to $\frac{2}{3}$?

A. $\frac{3}{6}$

B. $\frac{2}{6}$

C. $\frac{3}{2}$

D. $\frac{4}{6}$

Find the missing number.

$$\frac{3}{4} = \frac{6}{?}$$

Missing number: _____

True or false?

$$\frac{1}{3} = \frac{3}{6}$$

A. True

B. False

SET 39 • 4.NF.C.5

BACKWARD 2

A hundred is ten tens • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.1.a

How many tens are in 100?

Tens: _____

True or false?

7 tens is 70.

A. True

B. False

How many ones are in 4 tens?

A. 400

B. 40

C. 14

D. 4

How many tens are in 90?

Tens: _____

True or false?

10 tens make 1,000.

A. True

B. False

SET 39 • 4.NF.C.5

FORWARD 1

Add decimals to hundredths • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Add.

$$0.6 + 0.25$$

Sum: _____

True or false?

$$0.3 + 0.07 = 0.37$$

A. True

B. False

Add.

$$0.4 + 0.58$$

- A.** 0.62
- B.** 9.8
- C.** 0.98
- D.** 0.098

Add.

$$1.7 + 0.15$$

Sum: _____

True or false?

$$0.5 + 0.05 = 0.10$$

A. True

B. False

SET 39 • 4.NF.C.5

FORWARD 2

Write tenths and hundredths as percents • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write $\frac{6}{10}$ as a percent.

Percent: _____

True or false?

$$\frac{45}{100} = 45\%$$

A. True

B. False

$\frac{3}{10} + \frac{12}{100}$ is equal to which percent?

- A.** 15%
- B.** 33%
- C.** 42%
- D.** 312%

Write 8% as a fraction with a denominator of 100.

Fraction: _____

True or false?

$$\frac{9}{10} = 9\%$$

A. True

B. False

Write fractions with denominators 10 or 100 as decimals

MAIN	Grade 4 main question	4.NF.C.6
BACKWARD 1	Dimes, pennies, and dollars	2.MD.C.8
BACKWARD 2	Hundreds, tens, and ones	2.NBT.A.1
FORWARD 1	Write thousandths as decimals	5.NBT.A.3.a
FORWARD 2	Write decimals as percents	6.RP.A.3.c

SET 40 • 4.NF.C.6

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{47}{100}$ as a decimal.

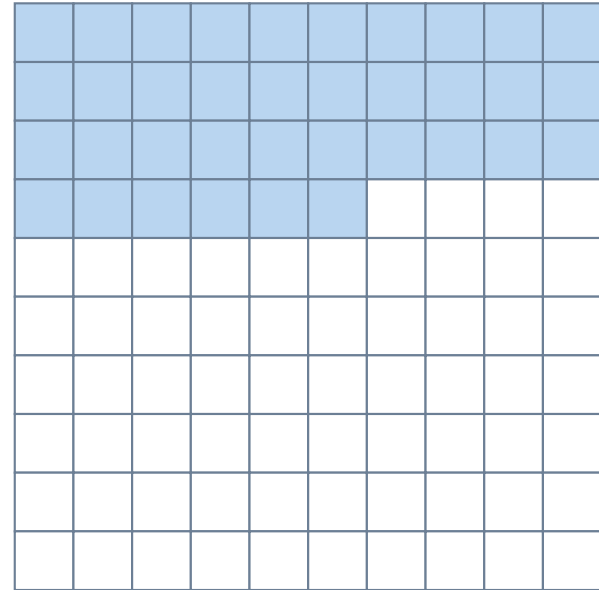
Decimal: _____

Write 0.8 as a fraction with a denominator of 10.

Fraction: _____

What decimal does the shaded part of the grid show?

- A.** 0.36
- B.** 3.6
- C.** 36
- D.** 0.036



True or false?

$$\frac{5}{100} = 0.5$$

A. True

B. False

A ribbon is $\frac{93}{100}$ meter long.

Write its length as a decimal.

Length: _____

SET 40 • 4.NF.C.6

BACKWARD 1

Dimes, pennies, and dollars • Grade 2 • Measurement & Data • 2.MD.C.8

How many cents are 3 dimes and 4 pennies?

Cents: _____

True or false?

10 dimes are worth 1 dollar.

A. True

B. False

Which coins make 25 cents?

- A.** 2 dimes and 5 pennies
- B.** 2 dimes and 2 pennies
- C.** 25 dimes
- D.** 1 dime and 5 pennies

How many pennies make 1 dollar?

Pennies: _____

True or false?

7 dimes are worth 7 cents.

A. True

B. False

SET 40 • 4.NF.C.6

BACKWARD 2

Hundreds, tens, and ones • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.1

How many hundreds, tens, and ones are in 245?

Answer: _____

True or false?

In 307, the 0 means 0 tens.

A. True

B. False

What is the value of the 5 in 452?

- A.** 45
- B.** 500
- C.** 5
- D.** 50

Write the number that is 6 tens and 2 ones.

Number: _____

True or false?

In 180, the 8 means 8 ones.

A. True

B. False

SET 40 • 4.NF.C.6

FORWARD 1

Write thousandths as decimals • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

Write $\frac{125}{1000}$ as a decimal.

Decimal: _____

True or false?

$$0.009 = \frac{9}{1000}$$

A. True

B. False

Which decimal is equal to $\frac{4}{1000}$?

- A.** 0.04
- B.** 0.004
- C.** 4.000
- D.** 0.4

Write 0.375 as a fraction with a denominator of 1,000.

Fraction: _____

True or false?

$$\frac{72}{1000} = 0.72$$

A. True

B. False

SET 40 • 4.NF.C.6

FORWARD 2

Write decimals as percents • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

Write 0.35 as a percent.

Percent: _____

True or false?

$$0.07 = 7\%$$

A. True

B. False

Which decimal is equal to 60%?

A. 0.6

B. 60

C. 0.06

D. 6.0

Write 120% as a decimal.

Decimal: _____

True or false?

$$0.5 = 5\%$$

A. True

B. False

Locate decimals to hundredths on a number line

MAIN	Grade 4 main question	4.NF.C.6
BACKWARD 1	Fractions on a number line	3.NF.A.2
BACKWARD 2	Whole numbers on a number line	2.MD.B.6
FORWARD 1	Round decimals using place value	5.NBT.A.4
FORWARD 2	Locate positive and negative decimals on a number line	6.NS.C.6.c

SET 41 • 4.NF.C.6

MAIN

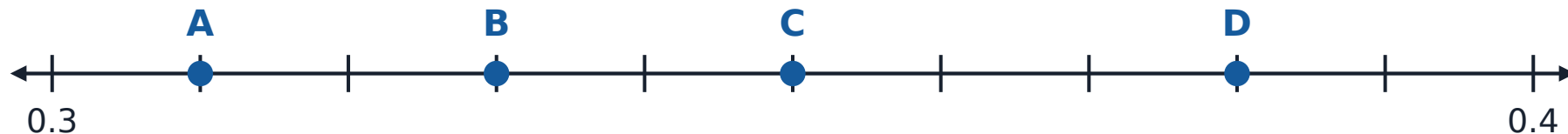
Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

What decimal is at point A?



A = _____

Which point is at 0.35?



A. Point A

B. Point B

C. Point C

D. Point D

Mark and label 0.4 and 0.85 on the number line.



True or false?

0.5 is halfway between 0 and 1 on a number line.

- A. True
- B. False

What decimal is at point B?



B = _____

SET 41 • 4.NF.C.6

BACKWARD 1

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point A?



A = _____

True or false?

$\frac{1}{2}$ is halfway between 0 and 1 on a number line.

A. True

B. False

Into how many equal parts is the space from 0 to 1 divided?



A. 2

B. 5

C. 3

D. 4

What fraction is at point B?



B = _____

True or false?

$\frac{3}{4}$ is to the left of $\frac{1}{4}$ on a number line.

A. True

B. False

SET 41 • 4.NF.C.6

BACKWARD 2

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point P?



P = _____

True or false?

On a number line, 12 is to the right of 9.

A. True

B. False

Which number is halfway between 20 and 30 on a number line?

A. 24

B. 25

C. 50

D. 15

What number is at point Q?



Q = _____

True or false?

The distance from 3 to 8 on a number line is 11.

A. True

B. False

SET 41 • 4.NF.C.6

FORWARD 1

Round decimals using place value • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 2.68 to the nearest tenth.

Rounded: _____

True or false?

On a number line, 4.17 is closer to 4.2 than to 4.1.

- A. True
- B. False

Which number is closest to 5?

A. 5.1

B. 4.8

C. 5.09

D. 4.96

Round 0.734 to the nearest hundredth.

Rounded: _____

True or false?

6.25 rounded to the nearest whole number is 7.

A. True

B. False

SET 41 • 4.NF.C.6

FORWARD 2

Locate positive and negative decimals on a number line • Grade 6 • The Number System • 6.NS.C.6.c

What number is at point A?



A = _____

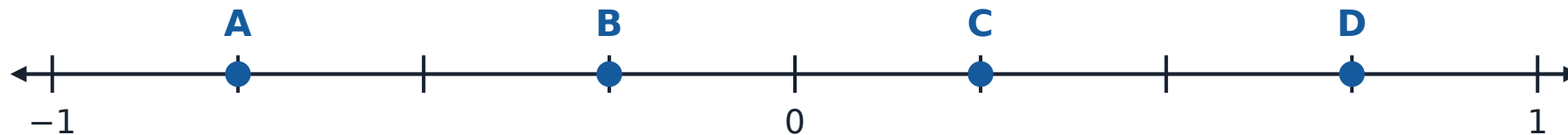
True or false?

-0.5 is halfway between -1 and 0 on a number line.

A. True

B. False

Which point is at -0.75 ?



A. Point A

B. Point B

C. Point C

D. Point D

Mark and label -1.2 and 0.8 on the number line.



True or false?

-1.25 is to the right of -1 on a number line.

A. True

B. False

Compare two decimals to hundredths and justify the comparison

MAIN	Grade 4 main question	4.NF.C.7
BACKWARD 1	Compare fractions with the same numerator or the same denomin...	3.NF.A.3.d
BACKWARD 2	Compare three-digit numbers	2.NBT.A.4
FORWARD 1	Compare decimals to thousandths	5.NBT.A.3.b
FORWARD 2	Write and interpret statements of order for decimals in context	6.NS.C.7.b

SET 42 • 4.NF.C.7

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.C.7

Write $>$, $=$, or $<$.

0.6 0.58

Symbol: _____

Which comparison is true?

A. $0.1 < 0.09$

B. $0.09 < 0.1$

C. $0.09 > 0.1$

D. $0.09 = 0.1$

True or false?

$$0.40 = 0.4$$

A. True

B. False

Compare 0.3 and 0.25. Write $>$, $=$, or $<$, and explain using place value or a model.

Comparison: _____

Explanation: _____

True or false?

Jo has 0.5 of a large pizza and Sam has 0.5 of a small pizza. They have the same amount of pizza.

A. True

B. False

SET 42 • 4.NF.C.7

BACKWARD 1

Compare fractions with the same numerator or the same denominator • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

Write $>$, $=$, or $<$.

$$\frac{5}{8} \text{ — } \frac{3}{8}$$

Symbol: _____

True or false?

$$\frac{1}{6} < \frac{1}{2}$$

A. True

B. False

Which fraction is least?

A. $\frac{2}{6}$

B. $\frac{2}{3}$

C. $\frac{2}{8}$

D. $\frac{2}{4}$

Write $>$, $=$, or $<$.

$$\frac{2}{4} \text{ — } \frac{2}{4}$$

Symbol: _____

True or false?

$$\frac{1}{8} > \frac{1}{4}$$

A. True

B. False

SET 42 • 4.NF.C.7

BACKWARD 2

Compare three-digit numbers • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

Write $>$, $=$, or $<$.

630 603

Symbol: _____

True or false?

$$458 < 485$$

A. True

B. False

Which number is least?

A. 901

B. 910

C. 291

D. 199

Write $>$, $=$, or $<$.

777 787

Symbol: _____

True or false?

$$350 > 530$$

A. True

B. False

SET 42 • 4.NF.C.7

FORWARD 1

Compare decimals to thousandths • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b

Write $>$, $=$, or $<$.

0.307 0.37

Symbol: _____

True or false?

$$4.5 > 4.485$$

A. True

B. False

Which decimal is least?

- A.** 0.1
- B.** 0.19
- C.** 0.109
- D.** 0.099

Write $>$, $=$, or $<$.

2.500 2.5

Symbol: _____

True or false?

$$0.62 < 0.602$$

A. True

B. False

SET 42 • 4.NF.C.7

FORWARD 2

Write and interpret statements of order for decimals in context • Grade 6 • The Number System • 6.NS.C.7.b

Write an inequality that compares -1.5°C and -0.5°C .

Inequality: _____

True or false?

A depth of -3.2 m is deeper than a depth of -2.3 m.

A. True

B. False

Which statement is true?

A. $0.3 < -0.8$

B. $-0.3 < -0.8$

C. $-0.8 < -0.3$

D. $-0.8 > -0.3$

Order from least to greatest.

-0.4 , 0.2 , -1.1

Order: _____

True or false?

$$-2.75 > -2.5$$

A. True

B. False

Know the relative sizes of measurement units within one system

MAIN	Grade 4 main question	4.MD.A.1
BACKWARD 1	Grams, kilograms, and liters	3.MD.A.2
BACKWARD 2	Measure with two different units	2.MD.A.2
FORWARD 1	Convert measurement units within one system	5.MD.A.1
FORWARD 2	Use ratio reasoning to convert measurement units	6.RP.A.3.d

SET 43 • 4.MD.A.1

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.1

Which unit is best for measuring the length of a river?

- A.** Centimeters
- B.** Kilometers
- C.** Grams
- D.** Liters

True or false?

1 kilogram is 1,000 times as heavy as 1 gram.

A. True

B. False

How many ounces are in 1 pound?

Ounces: _____

Which is heaviest?

- A.** 1 gram
- B.** 1 kilogram
- C.** 100 grams
- D.** 500 grams

True or false?

1 liter is less than 1 milliliter.

A. True

B. False

SET 43 • 4.MD.A.1

BACKWARD 1

Grams, kilograms, and liters • Grade 3 • Measurement & Data • 3.MD.A.2

Which object has a mass of about 1 kilogram?

- A.** A grape
- B.** A car
- C.** A large book
- D.** A paper clip

True or false?

A spoonful of water is about 1 liter.

A. True

B. False

Would you measure the mass of a pencil in grams or in kilograms?

Unit: _____

True or false?

A bucket can hold about 10 liters of water.

A. True

B. False

A bag of flour has a mass of 2 kg. A bag of sugar has a mass of 3 kg.

What is their total mass?

Mass: _____

SET 43 • 4.MD.A.1

BACKWARD 2

Measure with two different units • Grade 2 • Measurement & Data • 2.MD.A.2

True or false?

A desk measured in inches gives a larger number than the same desk measured in feet.

- A.** True
- B.** False

A table is 2 meters long.

About how many centimeters long is it?

- A.** 200
- B.** 20
- C.** 2,000
- D.** 2

A rope is 3 feet long.

Is it longer or shorter than 3 inches?

Answer: _____

True or false?

A book measured in centimeters gives a smaller number than the same book measured in meters.

A. True

B. False

Which is longer: 1 foot or 1 inch?

Answer: _____

SET 43 • 4.MD.A.1

FORWARD 1

Convert measurement units within one system • Grade 5 • Measurement & Data • 5.MD.A.1

Convert.

3.5 kilometers = ____ meters

Meters: _____

True or false?

2,500 grams = 2.5 kilograms

A. True

B. False

How many ounces are in $2\frac{1}{2}$ pounds?

- A.** 32
- B.** 20
- C.** 36
- D.** 40

Convert.

90 minutes = ____ hours

Hours: _____

True or false?

400 centimeters = 40 meters

A. True

B. False

SET 43 • 4.MD.A.1

FORWARD 2

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 12 inches in 1 foot.

How many feet are in 54 inches?

Feet: _____

True or false?

If 1 mile is about 1.6 km, then 5 miles is about 8 km.

A. True

B. False

There are 4 quarts in 1 gallon.

How many gallons are in 26 quarts?

A. 104

B. $6\frac{1}{2}$

C. 30

D. 22

A recipe uses 3 cups of milk for every 2 batches.

How many cups of milk are needed for 7 batches?

Cups: _____

True or false?

If 1 inch = 2.54 cm, then 10 inches = 2.54 cm.

- A. True
- B. False

Express a larger unit in terms of a smaller unit and record the pairs in a table

MAIN	Grade 4 main question	4.MD.A.1
BACKWARD 1	Multiply one-digit numbers by multiples of 10	3.NBT.A.3
BACKWARD 2	Patterns in a table	3.OA.D.9
FORWARD 1	Convert from a smaller unit to a larger unit	5.MD.A.1
FORWARD 2	Use tables of equivalent ratios to convert units	6.RP.A.3.d

SET 44 • 4.MD.A.1

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.1

Complete the table.

Meters	1	2	3	4
Centimeters	100	200	?	?

Missing values: _____

How many ounces are in 5 pounds?

Ounces: _____

A movie is 2 hours long.

How many minutes is that?

A. 200

B. 60

C. 120

D. 24

True or false?

7 kilograms = 700 grams

A. True

B. False

Complete the table.

Hours	1	2	3	4
Minutes	60	?	180	?

Missing values: _____

SET 44 • 4.MD.A.1

BACKWARD 1

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations
in Base Ten • 3.NBT.A.3

Multiply.

$$4 \times 60$$

Product: _____

True or false?

$$2 \times 50 = 100$$

A. True

B. False

$$8 \times 60 = ?$$

- A.** 480
- B.** 48
- C.** 4,800
- D.** 68

Multiply.

$$3 \times 70$$

Product: _____

True or false?

$$6 \times 60 = 3,600$$

A. True

B. False

SET 44 • 4.MD.A.1

BACKWARD 2

Patterns in a table • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

What comes next?

16, 32, 48, ____

Next: _____

True or false?

In 15, 30, 45, 60, each number is 15 more than the one before.

- A. True
- B. False

What is the rule for 10, 20, 30, 40?

- A.** Add 10
- B.** Add 1
- C.** Multiply by 2
- D.** Add 20

What comes next?

12, 24, 36, ____

Next: _____

True or false?

The pattern 5, 10, 20, 40 adds 5 each time.

A. True

B. False

SET 44 • 4.MD.A.1

FORWARD 1

Convert from a smaller unit to a larger unit • Grade 5 • Measurement & Data •
5.MD.A.1

Convert.

48 ounces = ____ pounds

Pounds: _____

True or false?

2,000 milliliters = 2 liters

A. True

B. False

Convert.

250 centimeters = ____ meters

A. 2,500

B. 0.25

C. 25

D. 2.5

Convert.

150 seconds = ____ minutes

Minutes: _____

True or false?

36 inches = 4 feet

A. True

B. False

SET 44 • 4.MD.A.1

FORWARD 2

Use tables of equivalent ratios to convert units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

Complete the table.

Yards	1	2	5
Feet	3	6	?

Missing value: _____

True or false?

A table with the pairs (1, 1,000), (2, 2,000), and (3, 3,000) converts kilograms to grams.

- A. True
- B. False

There are 16 ounces in 1 pound.

How many pounds are 40 ounces?

A. 640

B. $2\frac{1}{2}$

C. 24

D. 3

1 liter = 1,000 milliliters.

How many liters are 3,500 milliliters?

Liters: _____

True or false?

Because $1 \text{ meter} = 100 \text{ cm}$, $0.75 \text{ meter} = 7.5 \text{ cm}$.

A. True

B. False

Solve word problems about distances and intervals of time

MAIN	Grade 4 main question	4.MD.A.2
BACKWARD 1	Time intervals in minutes	3.MD.A.1
BACKWARD 2	Length word problems within 100	2.MD.B.5
FORWARD 1	Solve multistep problems that involve converting units	5.MD.A.1
FORWARD 2	Solve constant-speed problems	6.RP.A.3.b

SET 45 • 4.MD.A.2

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.2

A movie started at 3:45 p.m. and ended at 5:20 p.m.

How long was the movie?

Length: _____

Ty biked 2 km 350 m on Monday and 1 km 900 m on Tuesday.

How far did he bike in all? Give the answer in meters.

Meters: _____

A race is 3 km long. Ana has run 1,850 m.

How many more meters must she run?

A. 1,150 m

B. 4,850 m

C. 1,250 m

D. 2,150 m

True or false?

Soccer practice starts at 4:30 p.m. and lasts 1 hour 15 minutes. It ends at 5:45 p.m.

- A.** True
- B.** False

A bus trip takes 45 minutes each way.

How long does a round trip take? Write the answer in hours and minutes.

Time: _____

SET 45 • 4.MD.A.2

BACKWARD 1

Time intervals in minutes • Grade 3 • Measurement & Data • 3.MD.A.1

A class starts at 9:10 and ends at 9:55.

How many minutes long is it?

Minutes: _____

True or false?

From 2:30 to 3:15 is 45 minutes.

A. True

B. False

Jo started reading at 7:20 and read for 25 minutes.

What time did she stop?

A. 7:35

B. 7:45

C. 8:45

D. 7:25

A show begins at 4:05 and lasts 50 minutes.
What time does it end?

Time: _____

True or false?

From 10:40 to 11:00 is 40 minutes.

A. True

B. False

SET 45 • 4.MD.A.2

BACKWARD 2

Length word problems within 100 • Grade 2 • Measurement & Data • 2.MD.B.5

A ribbon is 64 cm long. Mia cuts off 28 cm.
How long is the ribbon now?

Length: _____

True or false?

A 45-inch board and a 37-inch board laid end to end are 82 inches long.

A. True

B. False

A path is 90 m long. Sam has walked 55 m.

How far is left?

- A.** 45 m
- B.** 35 m
- C.** 25 m
- D.** 145 m

A tree was 38 feet tall. It grew 6 feet.

How tall is it now?

Height: _____

True or false?

Cutting 17 cm from a 50-cm rope leaves 37 cm.

A. True

B. False

SET 45 • 4.MD.A.2

FORWARD 1

Solve multistep problems that involve converting units • Grade 5 •
Measurement & Data • 5.MD.A.1

A runner ran 1.5 km and then 800 m.
How many meters did she run in all?

Meters: _____

True or false?

A trip of 2 hours 30 minutes takes 150 minutes.

A. True

B. False

A 4-yard rope has 5 feet cut off.

How many feet are left?

A. 17 feet

B. 1 foot

C. 7 feet

D. 9 feet

A train ride lasts $2\frac{1}{4}$ hours.

How many minutes is that?

Minutes: _____

True or false?

$$3 \text{ km} - 1,200 \text{ m} = 2,800 \text{ m}$$

A. True

B. False

SET 45 • 4.MD.A.2

FORWARD 2

Solve constant-speed problems • Grade 6 • Ratios & Proportional Relationships
• 6.RP.A.3.b

A car travels 150 miles in 3 hours at a constant speed.
What is its speed in miles per hour?

Speed: _____

True or false?

Walking 6 km in 2 hours is a speed of 3 km per hour.

- A. True
- B. False

A cyclist rides 12 miles per hour.

How far does she ride in 2.5 hours?

- A.** 14.5 miles
- B.** 30 miles
- C.** 24 miles
- D.** 36 miles

A train goes 240 km in 4 hours.

At this rate, how long does it take to go 420 km?

Time: _____

True or false?

At 60 miles per hour, a car goes 90 miles in 2 hours.

A. True

B. False

Solve word problems about liquid volume, mass, and money

MAIN	Grade 4 main question	4.MD.A.2
BACKWARD 1	Mass and liquid volume word problems	3.MD.A.2
BACKWARD 2	Dollars and cents	2.MD.C.8
FORWARD 1	Solve money and measurement problems with decimal operations	5.NBT.B.7
FORWARD 2	Find and compare unit prices	6.RP.A.3.b

SET 46 • 4.MD.A.2

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.2

A jug holds 2 liters of juice. Ella pours 6 cups of 250 milliliters each.

How many milliliters are left in the jug?

Milliliters: _____

A box of apples has a mass of 3 kg 400 g. Another box has a mass of 2 kg 750 g.

What is their total mass in grams?

Grams: _____

Jo buys 3 notebooks that cost \$2.45 each.
How much does she spend?

- A.** \$7.35
- B.** \$6.35
- C.** \$5.35
- D.** \$7.45

True or false?

Lee pays for items that cost \$3.60 and \$4.15 with a \$10 bill. His change is \$3.25.

A. True

B. False

A cat has a mass of 4 kg. A dog has 6 times as much mass.

What is the dog's mass in grams?

Grams: _____

SET 46 • 4.MD.A.2

BACKWARD 1

Mass and liquid volume word problems • Grade 3 • Measurement & Data •
3.MD.A.2

A pot holds 8 liters. 3 liters are poured out.
How many liters are left?

Liters: _____

True or false?

Four 5-kg bags have a total mass of 20 kg.

A. True

B. False

45 liters of water are shared equally among 9 buckets.

How many liters go in each bucket?

A. 54

B. 9

C. 5

D. 36

A melon has a mass of 900 g. A pear has a mass of 200 g.

How much more mass does the melon have?

Grams: _____

True or false?

Two 35-liter tubs hold 60 liters in all.

A. True

B. False

SET 46 • 4.MD.A.2

BACKWARD 2

Dollars and cents • Grade 2 • Measurement & Data • 2.MD.C.8

How much money is 2 quarters, 1 dime, and 3 pennies?

Cents: _____

True or false?

3 quarters are worth 75 cents.

A. True

B. False

Ben has 2 dollar bills and 4 dimes.

How much money does he have?

A. \$2.04

B. \$2.40

C. \$6.00

D. \$2.10

A toy costs 85 cents. Ann pays with 1 dollar.
How much change does she get?

Change: _____

True or false?

5 nickels and 2 dimes are worth 50 cents.

A. True

B. False

SET 46 • 4.MD.A.2

FORWARD 1

Solve money and measurement problems with decimal operations • Grade 5 •
Number & Operations in Base Ten • 5.NBT.B.7

Pens cost \$1.25 each.

How much do 8 pens cost?

Cost: _____

True or false?

A \$20 bill pays for 3 shirts that cost \$5.99 each.
The change is \$2.03.

- A.** True
- B.** False

A 2.4-liter bottle is shared equally among 6 cups.

How many liters go in each cup?

A. 1.4

B. 0.4

C. 4

D. 0.04

Grapes cost \$2.80 per pound.

How much do 3.5 pounds cost?

Cost: _____

True or false?

A 1.5-kg bag and a 0.75-kg bag have a total mass of 1.125 kg.

- A. True
- B. False

SET 46 • 4.MD.A.2

FORWARD 2

Find and compare unit prices • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

A 12-pack of water costs \$4.80.

What is the price of 1 bottle?

Price: _____

True or false?

4 kg of rice for \$6 costs \$1.50 per kilogram.

A. True

B. False

Which is the best buy?

- A.** 2 liters for \$3.20
- B.** 3 liters for \$4.50
- C.** 4 liters for \$6.40
- D.** 1 liter for \$1.70

5 pounds of apples cost \$8.75.

How much does 1 pound cost?

Cost: _____

True or false?

A pack of 6 juice boxes for \$3.60 costs \$0.70 per box.

A. True

B. False

Solve measurement problems with simple fractions or decimals, using number line diagrams

MAIN	Grade 4 main question	4.MD.A.2
BACKWARD 1	Solve time problems with a number line	3.MD.A.1
BACKWARD 2	Fractions on a number line	3.NF.A.2
FORWARD 1	Solve measurement problems by adding and subtracting fractions...	5.NF.A.2
FORWARD 2	Find distances between points with the same first or second coor...	6.NS.C.8

SET 47 • 4.MD.A.2

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.2

A board is 6 feet long. Jo cuts off $\frac{3}{4}$ foot.

How long is the board now?

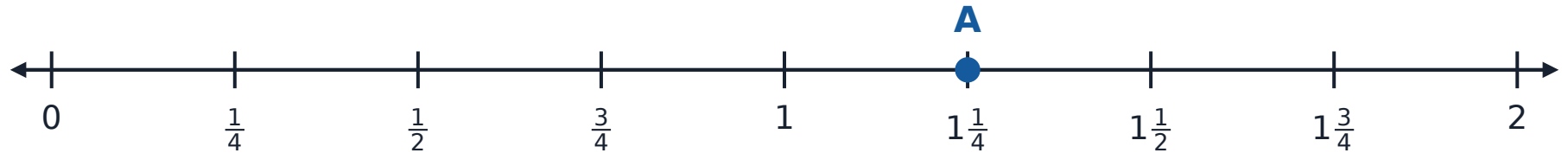
Length: _____

Ana ran 0.5 km before lunch and 0.25 km after lunch.
How far did she run in all?

Distance: _____

The number line shows a 2-mile hike. Lu has hiked to point A.

How much farther must she hike to reach the end?



- A.** $\frac{1}{4}$ mile **B.** 2 miles **C.** $\frac{3}{4}$ mile **D.** $1\frac{1}{4}$ miles

True or false?

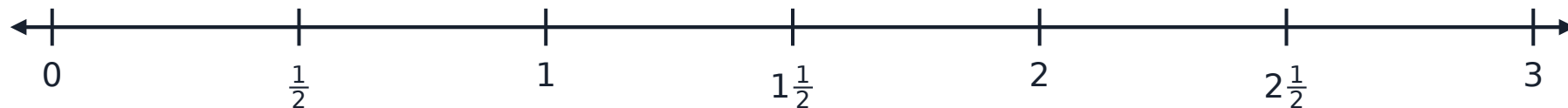
A jug has 2 liters of water. Lee pours out $\frac{1}{4}$ liter.
 $1\frac{3}{4}$ liters are left.

A. True

B. False

A rope is 3 meters long. Use the number line to show $\frac{1}{2}$ meter being cut off.

How much rope is left?



Rope left: _____

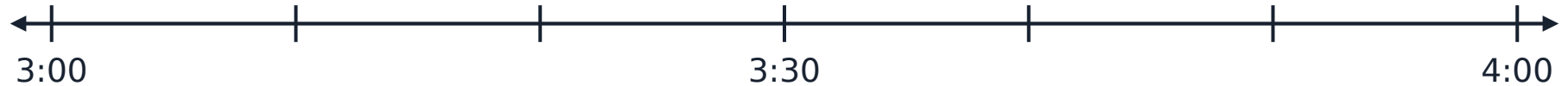
SET 47 • 4.MD.A.2

BACKWARD 1

Solve time problems with a number line • Grade 3 • Measurement & Data •
3.MD.A.1

Ty starts at 3:00 and works for 40 minutes. Use the number line.

What time does he finish?



Time: _____

True or false?

On a number line from 1:00 to 2:00, 1:30 is halfway.

A. True

B. False

A game started at 6:15 and lasted 35 minutes.

What time did it end?

A. 6:40

B. 6:35

C. 7:50

D. 6:50

How many minutes are there from 8:20 to 9:00?

Minutes: _____

True or false?

From 4:45 to 5:05 is 30 minutes.

A. True

B. False

SET 47 • 4.MD.A.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point A?



A = _____

True or false?

$\frac{3}{3}$ and 1 are at the same point on a number line.

A. True

B. False

Which fraction is closest to 0?

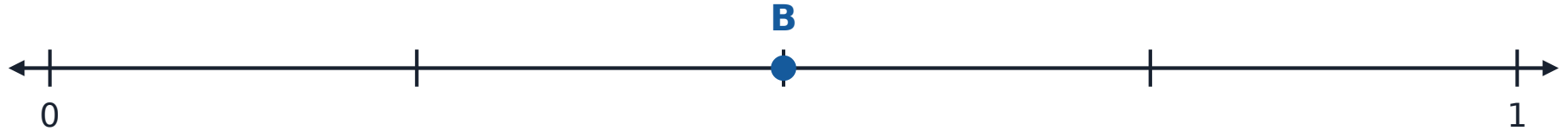
A. $\frac{1}{8}$

B. $\frac{3}{4}$

C. $\frac{1}{2}$

D. $\frac{7}{8}$

What fraction is at point B?



B = _____

True or false?

$\frac{1}{4}$ is to the right of $\frac{3}{4}$ on a number line.

A. True

B. False

SET 47 • 4.MD.A.2

FORWARD 1

Solve measurement problems by adding and subtracting fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

A board is 5 feet long. $\frac{2}{3}$ foot is cut off.

How long is the board now?

Length: _____

True or false?

A jug had $1\frac{1}{2}$ liters. $\frac{3}{4}$ liter was used. $\frac{3}{4}$ liter is left.

A. True

B. False

Ann walked $\frac{1}{2}$ km and then $\frac{3}{5}$ km.

How far did she walk in all?

A. $1\frac{1}{5}$ km

B. $1\frac{1}{10}$ km

C. $\frac{4}{10}$ km

D. $\frac{4}{7}$ km

A rope is $4\frac{1}{2}$ m long. A piece $1\frac{2}{3}$ m long is cut off.

How much rope is left?

Length: _____

True or false?

$$\frac{3}{8} \text{ kg} + \frac{1}{4} \text{ kg} = \frac{4}{12} \text{ kg}$$

A. True

B. False

SET 47 • 4.MD.A.2

FORWARD 2

Find distances between points with the same first or second coordinate • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(2, 5)$ and $(2, -3)$.

Distance: _____

True or false?

The distance between $(-4, 1)$ and $(3, 1)$ is 7 units.

A. True

B. False

On a map grid, a school is at $(-2, 4)$ and a park is at $(5, 4)$.

How many units apart are they?

A. 6

B. 7

C. 9

D. 3

Find the distance between $(-6, -2)$ and $(-6, 3)$.

Distance: _____

True or false?

The distance between $(0, -5)$ and $(0, 5)$ is 0 units.

A. True

B. False

Apply the area and perimeter formulas for rectangles

MAIN	Grade 4 main question	4.MD.A.3
BACKWARD 1	Area as length times width	3.MD.C.7.b
BACKWARD 2	Perimeter of polygons	3.MD.D.8
FORWARD 1	Find the area of rectangles with fractional side lengths	5.NF.B.4.b
FORWARD 2	Find the area of triangles and parallelograms	6.G.A.1

SET 48 • 4.MD.A.3

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.3

Find the area of the rectangle.

Area: _____



Find the perimeter of the rectangle.

Perimeter: _____



A rectangular rug is 8 feet by 6 feet.

What is its area?

- A.** 96 square feet
- B.** 14 square feet
- C.** 48 square feet
- D.** 28 square feet

True or false?

A square with sides of 7 cm has a perimeter of 49 cm.

A. True

B. False

A garden is 15 m long and 8 m wide.
What are its area and its perimeter?

Area: _____

Perimeter: _____

SET 48 • 4.MD.A.3

BACKWARD 1

Area as length times width • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle is 6 units by 7 units.

What is its area?

Area: _____

True or false?

A 5-by-9 rectangle has an area of 45 square units.

A. True

B. False

Which equation gives the area of a 4-by-8 rectangle?

A. $8 - 4 = 4$

B. $4 + 8 = 12$

C. $2 \times (4 + 8) = 24$

D. $4 \times 8 = 32$

Find the area of the rectangle.

Area: _____



True or false?

A 3-by-9 rectangle has an area of 12 square units.

A. True

B. False

SET 48 • 4.MD.A.3

BACKWARD 2

Perimeter of polygons • Grade 3 • Measurement & Data • 3.MD.D.8

A triangle has sides of 5 cm, 7 cm, and 9 cm.
What is its perimeter?

Perimeter: _____

True or false?

A square with sides of 4 m has a perimeter of 16 m.

A. True

B. False

What is the perimeter of a rectangle that is 6 ft by 2 ft?

- A.** 12 feet
- B.** 8 feet
- C.** 14 feet
- D.** 16 feet

A pentagon has five sides that are each 3 inches long.
What is its perimeter?

Perimeter: _____

True or false?

A rectangle that is 5 units by 3 units has a perimeter of 15 units.

- A. True
- B. False

SET 48 • 4.MD.A.3

FORWARD 1

Find the area of rectangles with fractional side lengths • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.b

Find the area of a rectangle that is $\frac{3}{4}$ m by $\frac{2}{3}$ m.

Area: _____

True or false?

A rectangle that is $2\frac{1}{2}$ ft by 4 ft has an area of 10 square feet.

- A.** True
- B.** False

A square has sides that are $\frac{1}{2}$ inch long.

What is its area?

A. 2 square inches

B. 1 square inch

C. $\frac{1}{4}$ square inch

D. $\frac{1}{2}$ square inch

Find the area of a rectangle that is $1\frac{1}{2}$ yd by $\frac{2}{3}$ yd.

Area: _____

True or false?

A $\frac{1}{3}$ -by- $\frac{1}{3}$ square has an area of $\frac{2}{3}$ square unit.

A. True

B. False

SET 48 • 4.MD.A.3

FORWARD 2

Find the area of triangles and parallelograms • Grade 6 • Geometry • 6.G.A.1

A triangle has a base of 10 cm and a height of 6 cm.
What is its area?

Area: _____

True or false?

A parallelogram with a base of 8 m and a height of 5 m has an area of 40 square meters.

A. True

B. False

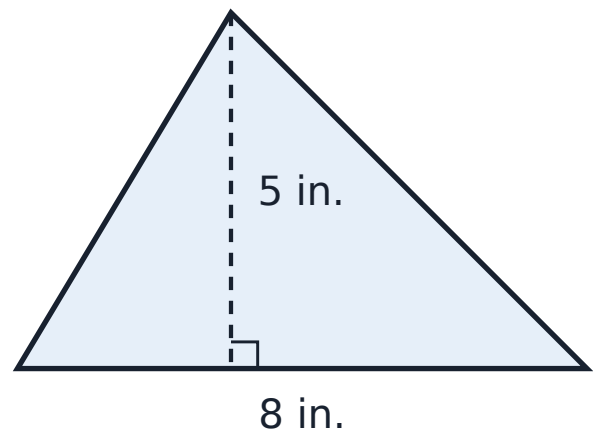
A right triangle has legs that are 4 ft and 9 ft long.

What is its area?

- A.** 26 square feet
- B.** 18 square feet
- C.** 36 square feet
- D.** 13 square feet

Find the area of the triangle.

Area: _____



True or false?

A triangle with a base of 6 units and a height of 4 units has an area of 24 square units.

- A. True
- B. False

Find an unknown side length from the area or the perimeter of a rectangle

MAIN	Grade 4 main question	4.MD.A.3
BACKWARD 1	Find an unknown side from the perimeter	3.MD.D.8
BACKWARD 2	Find the unknown factor	3.OA.A.4
FORWARD 1	Find a missing dimension when the volume is known	5.MD.C.5.b
FORWARD 2	Solve equations of the form $px = q$ for a missing measurement	6.EE.B.7

SET 49 • 4.MD.A.3

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has an area of 72 square feet and a length of 9 feet.

What is its width?

Width: _____

A rectangle has a perimeter of 30 cm and a length of 10 cm.

What is its width?

Width: _____

A rectangular patio has an area of 96 square meters.
One side is 12 m long.

How long is the other side?

A. 36 m

B. 108 m

C. 84 m

D. 8 m

True or false?

A square has a perimeter of 36 inches, so each side is 9 inches long.

A. True

B. False

A rectangle has an area of 54 square units and a width of 6 units.

Write an equation with a letter for the length. Then solve it.

Equation: _____

Length: _____

SET 49 • 4.MD.A.3

BACKWARD 1

Find an unknown side from the perimeter • Grade 3 • Measurement & Data •
3.MD.D.8

A triangle has a perimeter of 20 cm. Two sides are 6 cm and 8 cm long.

How long is the third side?

Length: _____

True or false?

A square with a perimeter of 12 m has sides that are 3 m long.

A. True

B. False

A rectangle has a perimeter of 14 ft and a length of 5 ft.
What is its width?

- A.** 7 ft
- B.** 4 ft
- C.** 2 ft
- D.** 9 ft

A rectangle has a perimeter of 18 in. and a width of 4 in.
What is its length?

Length: _____

True or false?

A square with a perimeter of 20 units has sides that are 10 units long.

- A. True
- B. False

SET 49 • 4.MD.A.3

BACKWARD 2

Find the unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

Find the unknown number.

$$8 \times ? = 64$$

Unknown: _____

True or false?

If $9 \times w = 45$, then $w = 5$.

A. True

B. False

$$? \times 7 = 56$$

A. 7

B. 8

C. 49

D. 63

Find the unknown number.

$$6 \times ? = 42$$

Unknown: _____

True or false?

If $4 \times n = 28$, then $n = 6$.

A. True

B. False

SET 49 • 4.MD.A.3

FORWARD 1

Find a missing dimension when the volume is known • Grade 5 • Measurement & Data • 5.MD.C.5.b

A box has a volume of 60 cubic inches. It is 5 in. long and 3 in. wide.

How tall is it?

Height: _____

True or false?

A prism with a volume of 48 cubic cm and a base area of 12 square cm is 4 cm tall.

A. True

B. False

A tank holds 120 cubic feet. Its base is 6 ft by 5 ft.

How deep is the tank?

- A.** 20 feet
- B.** 4 feet
- C.** 10 feet
- D.** 109 feet

A prism has a volume of 90 cubic meters and a height of 5 m.

What is the area of its base?

Base area: _____

True or false?

A box that is 4 units by 3 units at the base and has a volume of 36 cubic units is 4 units tall.

A. True

B. False

SET 49 • 4.MD.A.3

FORWARD 2

Solve equations of the form $px = q$ for a missing measurement • Grade 6 •
Expressions & Equations • 6.EE.B.7

A rectangle has an area of 7.5 square meters and a length of 3 m.

Write and solve an equation for its width w .

Equation: _____

$W =$ _____

True or false?

A triangle has an area of 24 square units and a base of 8 units. Solving $4h = 24$ gives a height of 6 units.

- A. True
- B. False

Solve for b.

$$12b = 84$$

A. 1,008

B. 7

C. 72

D. 96

Solve for x.

$$\frac{1}{2}x = 9$$

x = _____

True or false?

The solution of $4w = 30$ is $w = 26$.

A. True

B. False

Make a line plot of measurements in fractions of a unit ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$)

MAIN	Grade 4 main question	4.MD.B.4
BACKWARD 1	Read a line plot with halves and fourths	3.MD.B.4
BACKWARD 2	Line plots of whole-number lengths	2.MD.D.9
FORWARD 1	Make line plots and use operations on the data	5.MD.B.2
FORWARD 2	Display numerical data in dot plots	6.SP.B.4

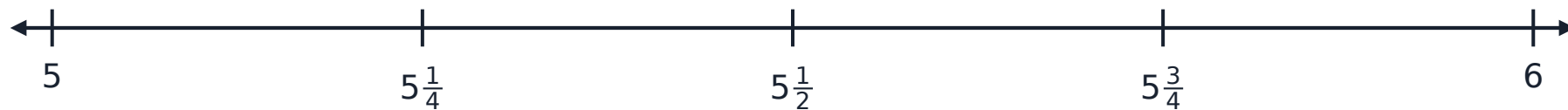
SET 50 • 4.MD.B.4

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.B.4

Pencil lengths (inches): $5\frac{1}{4}$, $5\frac{1}{2}$, $5\frac{1}{4}$, 6, $5\frac{3}{4}$, $5\frac{1}{2}$, $5\frac{1}{4}$

Make a line plot of the data.



Seedling heights (inches): $\frac{1}{8}$, $\frac{3}{8}$, $\frac{3}{8}$, $\frac{1}{2}$, $\frac{5}{8}$, $\frac{3}{8}$, $\frac{1}{8}$

Make a line plot of the data.



Shell lengths (inches): $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{2}$, 1, $\frac{3}{4}$, $\frac{1}{2}$

In a line plot of the data, how many X's go above $\frac{1}{2}$?

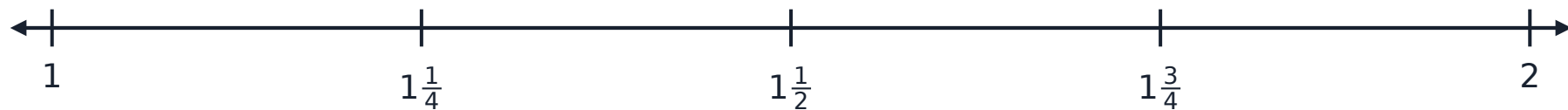
X's: _____

Which number line is best for a line plot of $\frac{1}{8}$, $\frac{3}{8}$, $\frac{5}{8}$, and $\frac{7}{8}$ inch?

- A.** A number line from 1 to 2 marked in fourths
- B.** A number line from 0 to 1 marked in halves
- C.** A number line from 0 to 1 marked in eighths
- D.** A number line from 0 to 8 marked in ones

Ribbon pieces (yards): $1\frac{1}{2}$, 2, $1\frac{1}{4}$, $1\frac{1}{2}$, $1\frac{3}{4}$, $1\frac{1}{2}$

Make a line plot of the data.



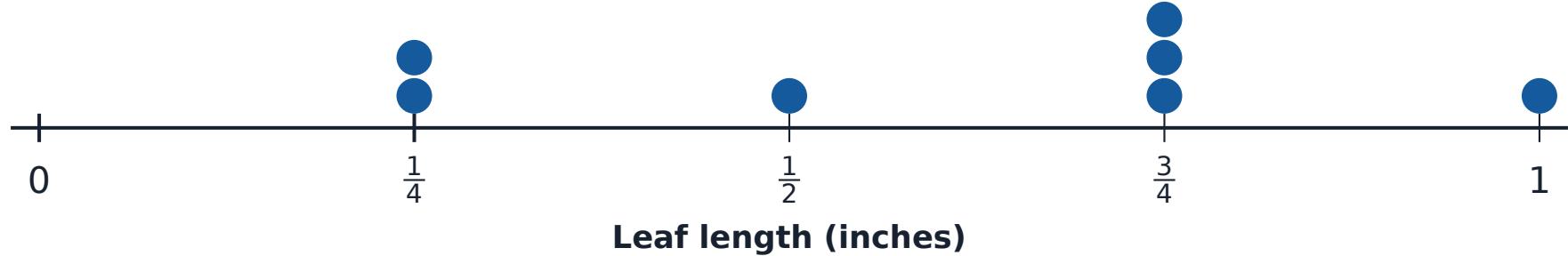
SET 50 • 4.MD.B.4

BACKWARD 1

Read a line plot with halves and fourths • Grade 3 • Measurement & Data •
3.MD.B.4

The line plot shows leaf lengths.

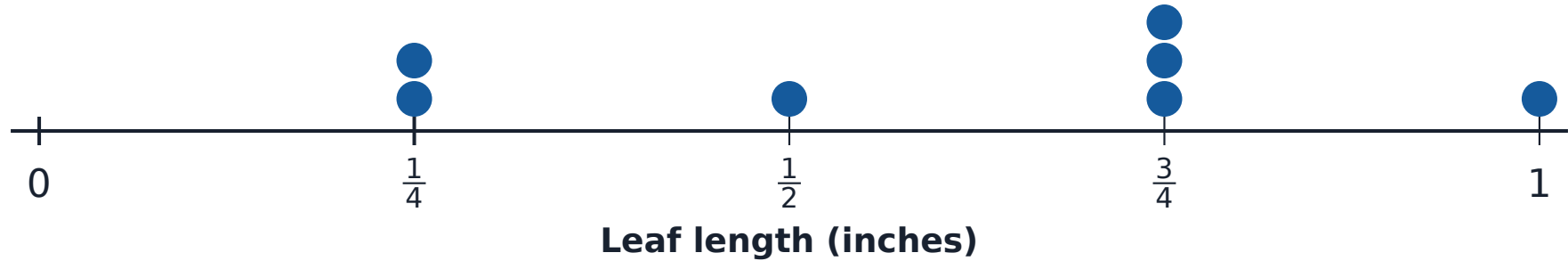
How many leaves are $\frac{3}{4}$ inch long?



Leaves: _____

True or false?

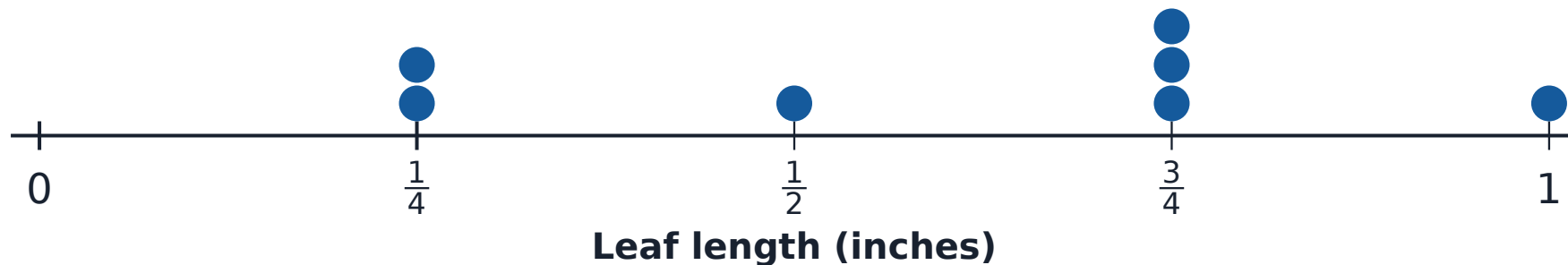
Two leaves are $\frac{1}{4}$ inch long.



A. True

B. False

Which leaf length is most common?



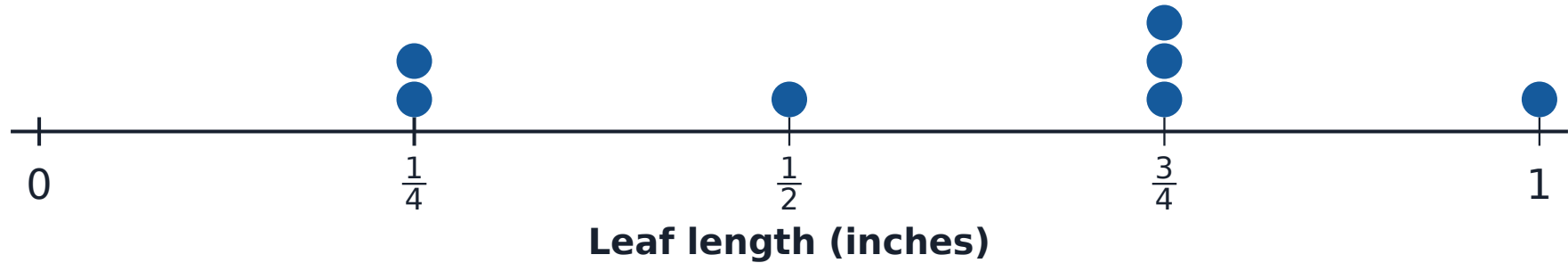
A. $\frac{1}{4}$ inch

B. $\frac{1}{2}$ inch

C. $\frac{3}{4}$ inch

D. 1 inch

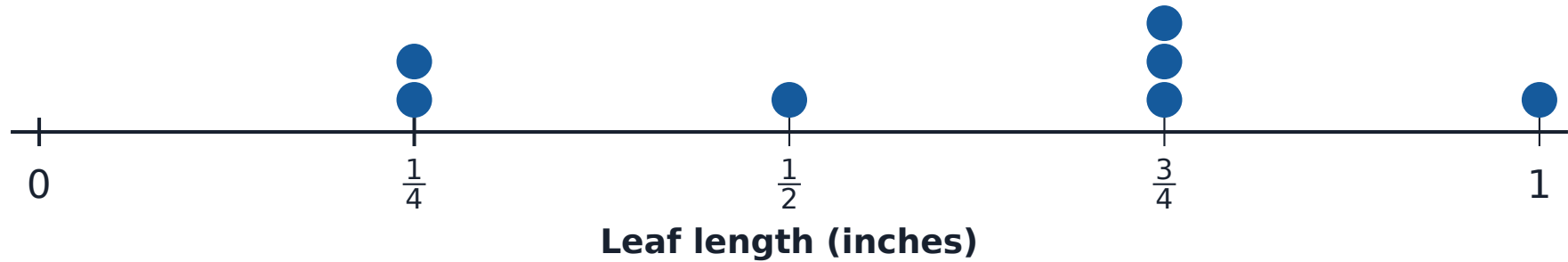
How many leaves were measured?



Leaves: _____

True or false?

No leaf is 1 inch long.



A. True

B. False

SET 50 • 4.MD.B.4

BACKWARD 2

Line plots of whole-number lengths • Grade 2 • Measurement & Data •
2.MD.D.9

Lengths (cm): 4, 6, 6, 7, 6

In a line plot of the data, how many X's go above 6?

X's: _____

True or false?

A line plot shows each measurement as a mark above a number line.

- A.** True
- B.** False

Which number line fits a line plot of 2, 3, 3, and 8?

- A.** A number line from 10 to 20
- B.** A number line from 0 to 3
- C.** A number line from 0 to 10
- D.** A number line from 4 to 6

Lengths (inches): 3, 5, 5, 6

Make a line plot of the data.



True or false?

A line plot of 1, 2, 2, 2 has 2 X's above 2.

- A. True
- B. False

SET 50 • 4.MD.B.4

FORWARD 1

Make line plots and use operations on the data • Grade 5 • Measurement & Data • 5.MD.B.2

Water in 4 cups (liters): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$

If the water were shared equally among the cups, how much would each cup hold?

Liters: _____

True or false?

The total of $\frac{1}{8}$, $\frac{3}{8}$, and $\frac{1}{2}$ inch is 1 inch.

A. True

B. False

A line plot shows 3 X's at $\frac{1}{4}$ pound and 2 X's at $\frac{1}{2}$ pound.

What is the total weight?

A. $1\frac{3}{4}$ pounds

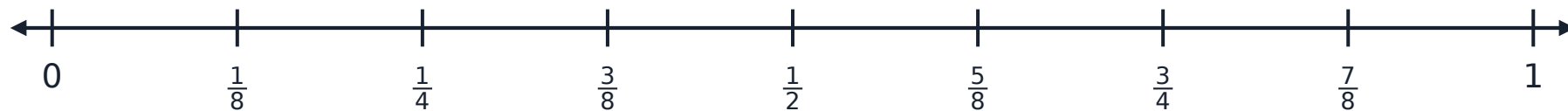
B. $\frac{3}{4}$ pound

C. $1\frac{1}{4}$ pounds

D. 5 pounds

Rain (inches): $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{2}$

Make a line plot of the data.



True or false?

The difference between the largest and smallest of $\frac{1}{8}$, $\frac{3}{8}$, and $\frac{7}{8}$ is $\frac{5}{8}$.

A. True

B. False

SET 50 • 4.MD.B.4

FORWARD 2

Display numerical data in dot plots • Grade 6 • Statistics & Probability •
6.SP.B.4

Number of pets: 0, 1, 1, 2, 2, 2, 3, 5

Make a dot plot of the data.



True or false?

A histogram groups data values into intervals.

A. True

B. False

Which display shows every individual data value?

- A.** A histogram
- B.** A table of intervals
- C.** A dot plot
- D.** A box plot

Data: 12, 15, 15, 18, 20

In a dot plot of the data, how many dots go above 15?

Dots: _____

True or false?

A dot plot cannot show data with repeated values.

- A. True
- B. False

Solve addition and subtraction problems using fraction data in a line plot

MAIN	Grade 4 main question	4.MD.B.4
BACKWARD 1	Read a line plot of lengths in fourths of an inch	3.MD.B.4
BACKWARD 2	Equivalent halves, fourths, and eighths	3.NF.A.3.b
FORWARD 1	Use operations on fractions to solve line plot problems	5.MD.B.2
FORWARD 2	Find the mean of a data set	6.SP.B.5.c

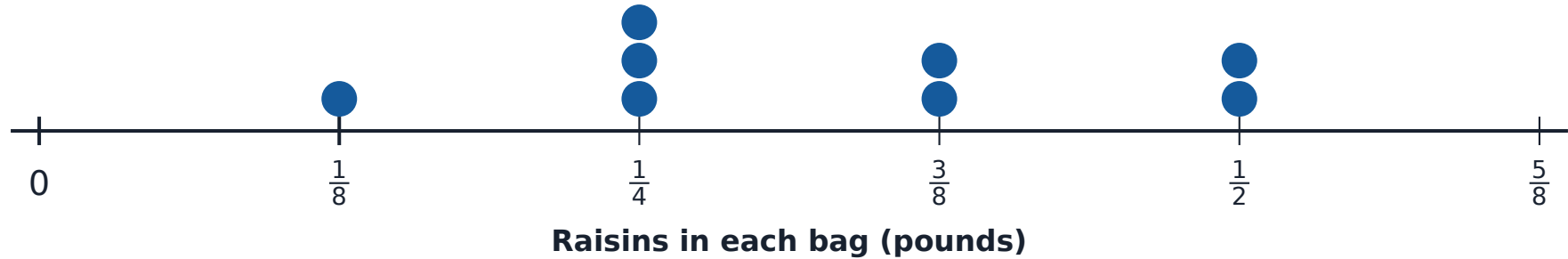
SET 51 • 4.MD.B.4

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows the raisins in some bags.

What is the difference between the heaviest and the lightest bag?



Difference: _____

What is the total weight of the bags that weigh $\frac{1}{4}$ pound each?



Total: _____

How many bags weigh more than $\frac{1}{4}$ pound?



A. 2

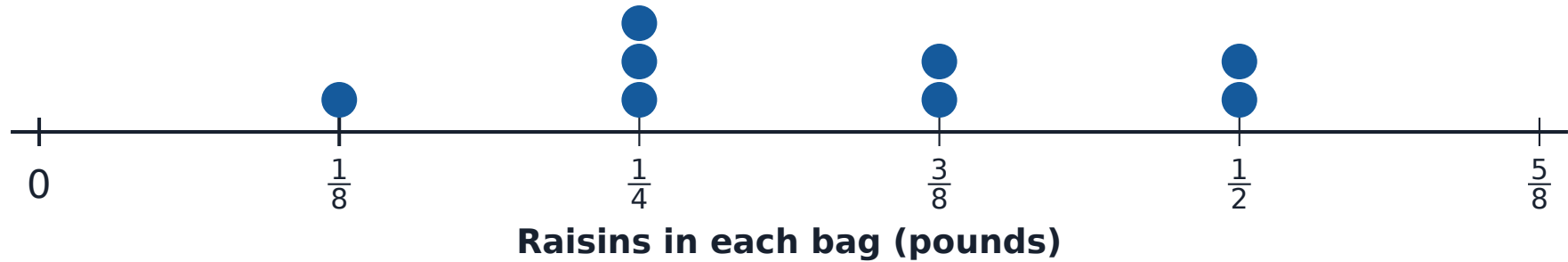
B. 3

C. 5

D. 4

True or false?

The two $\frac{1}{2}$ -pound bags weigh 1 pound in all.



A. True

B. False

What is the total weight of all the bags?



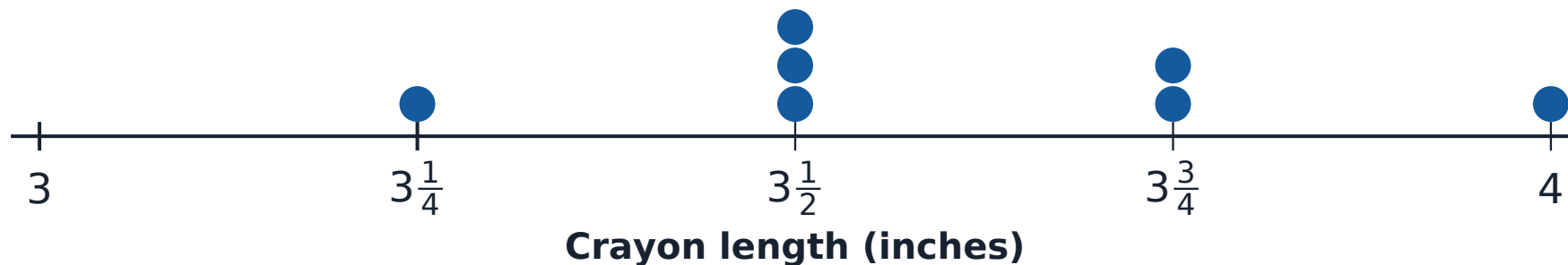
Total: _____

SET 51 • 4.MD.B.4

BACKWARD 1

Read a line plot of lengths in fourths of an inch • Grade 3 • Measurement & Data • 3.MD.B.4

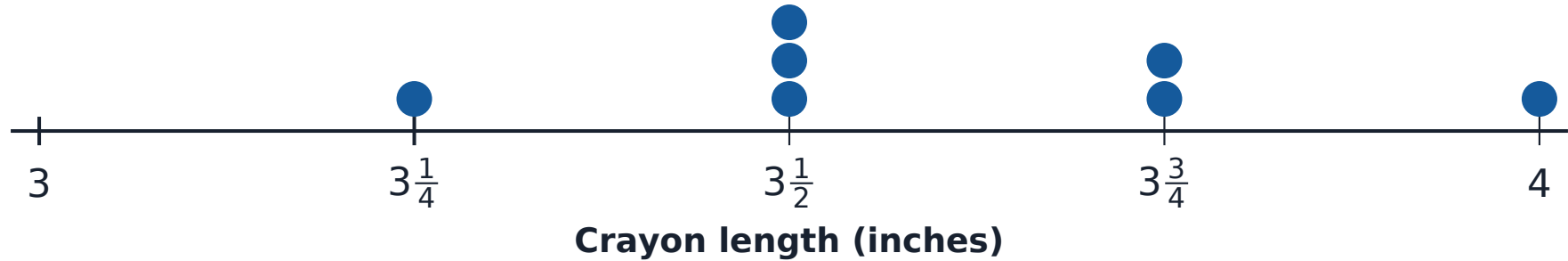
How many crayons are $3\frac{1}{2}$ inches long?



Crayons: _____

True or false?

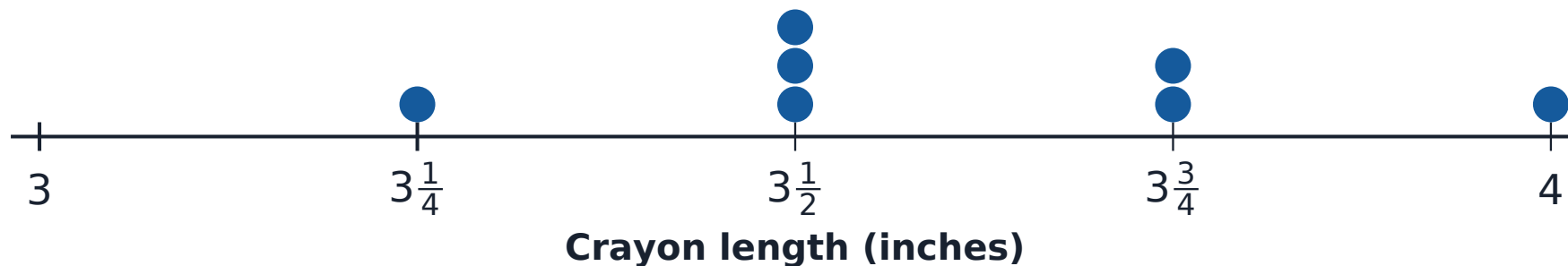
One crayon is 4 inches long.



A. True

B. False

Which length do exactly 2 crayons have?



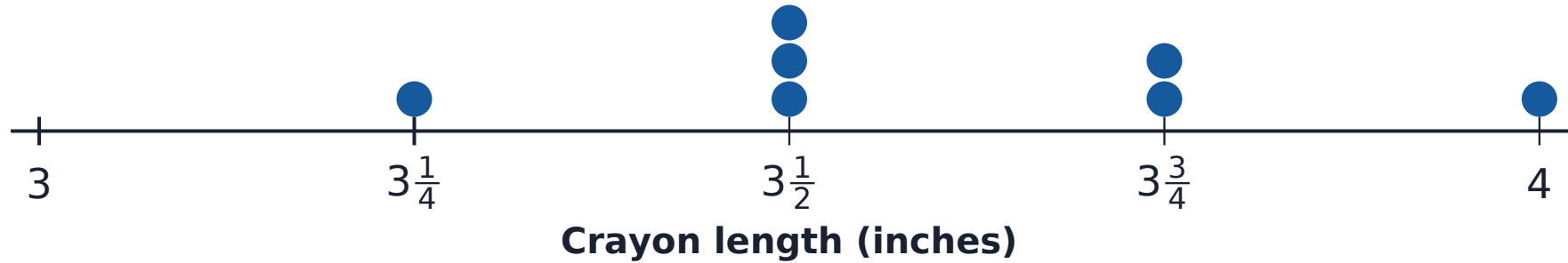
A. $3\frac{1}{2}$
inches

B. 4 inches

C. $3\frac{1}{4}$
inches

D. $3\frac{3}{4}$
inches

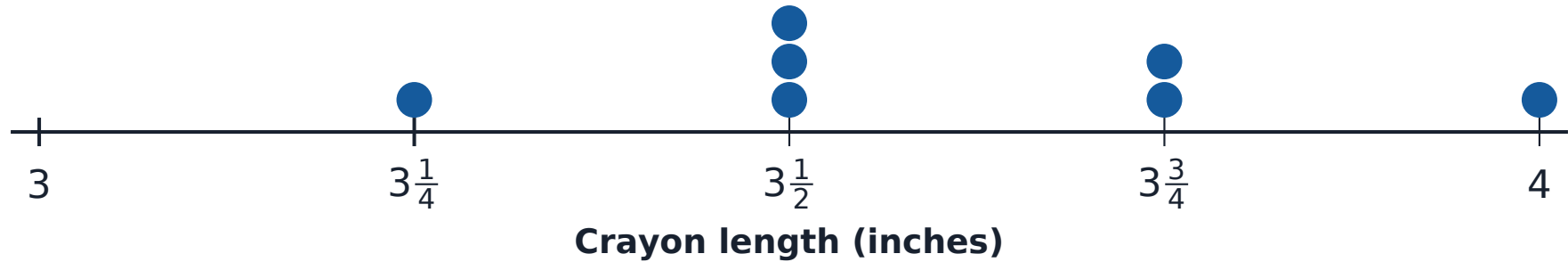
How many crayons were measured?



Crayons: _____

True or false?

The shortest crayon is 3 inches long.



A. True

B. False

SET 51 • 4.MD.B.4

BACKWARD 2

Equivalent halves, fourths, and eighths • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.b

Write $\frac{1}{2}$ as eighths.

Fraction: _____

True or false?

$$\frac{1}{4} = \frac{2}{8}$$

A. True

B. False

Which fraction is equal to $\frac{3}{4}$?

A. $\frac{5}{8}$

B. $\frac{6}{8}$

C. $\frac{4}{8}$

D. $\frac{3}{8}$

Find the missing number.

$$\frac{1}{2} = \frac{?}{4}$$

Missing number: _____

True or false?

$$\frac{2}{8} = \frac{1}{2}$$

A. True

B. False

SET 51 • 4.MD.B.4

FORWARD 1

Use operations on fractions to solve line plot problems • Grade 5 •
Measurement & Data • 5.MD.B.2

Juice in 3 glasses (cups): $\frac{1}{2}$, $\frac{3}{4}$, $1\frac{1}{4}$

If the juice were shared equally, how much would each glass hold?

Cups: _____

True or false?

Water in beakers (liters): $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$. The total is $\frac{3}{4}$ liter.

- A. True
- B. False

Ribbons (yards): $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{4}$

What is the total length?

A. $1\frac{3}{4}$ yards

B. $2\frac{1}{4}$ yards

C. $1\frac{1}{2}$ yards

D. 2 yards

Rock masses (kg): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$

If the total mass were shared equally by the 4 rocks, what would each rock's mass be?

Mass: _____

True or false?

The total of $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{8}$ is $\frac{3}{14}$.

A. True

B. False

SET 51 • 4.MD.B.4

FORWARD 2

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

4, 6, 8, 10

Mean: _____

True or false?

The mean of 2, 4, and 9 is 5.

A. True

B. False

Find the mean.

$$\frac{1}{2}, \frac{1}{2}, 1, 2$$

A. 4

B. $\frac{1}{2}$

C. 2

D. 1

Five plants grew 3, 5, 5, 6, and 6 cm.

What is the mean growth?

Mean: _____

True or false?

The mean of 10, 10, and 40 is 10.

A. True

B. False

An angle is measured as a fraction of a full circle; a one-degree angle is $\frac{1}{360}$ of a circle

MAIN	Grade 4 main question	4.MD.C.5.a
BACKWARD 1	Equal parts of a shape	3.G.A.2
BACKWARD 2	Turns of the minute hand on a clock	3.MD.A.1
FORWARD 1	Classify triangles by their angles	5.G.B.4
FORWARD 2	Find a percent of a full turn — nearest related (no direct Grade 6 s...	6.RP.A.3.c

SET 52 • 4.MD.C.5.a

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.5.a

How many one-degree angles make a full circle?

Answer: _____

True or false?

An angle that turns through $\frac{1}{4}$ of a full circle measures 90° .

- A. True
- B. False

What fraction of a full circle does a 180° angle turn through?

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. $\frac{1}{180}$

D. $\frac{1}{4}$

An angle turns through $\frac{1}{6}$ of a full circle.

What is its measure?

Measure: _____

True or false?

An angle that turns through $\frac{1}{3}$ of a full circle measures 30° .

- A. True
- B. False

SET 52 • 4.MD.C.5.a

BACKWARD 1

Equal parts of a shape • Grade 3 • Geometry • 3.G.A.2

A circle is cut into 4 equal parts.

What fraction of the circle is 1 part?

Fraction: _____

True or false?

A circle cut into 2 equal parts is cut into halves.

A. True

B. False

A circle is cut into 6 equal parts.

What fraction of the circle is each part?

A. $\frac{1}{3}$

B. $\frac{5}{6}$

C. $\frac{6}{1}$

D. $\frac{1}{6}$

Each part of a circle is $\frac{1}{8}$ of the circle.

Into how many equal parts is the circle cut?

Parts: _____

True or false?

A circle cut into 3 parts of different sizes is cut into thirds.

A. True

B. False

SET 52 • 4.MD.C.5.a

BACKWARD 2

Turns of the minute hand on a clock • Grade 3 • Measurement & Data •
3.MD.A.1